

Sensitivity, Causality, and Repair Dissociate: A Layer-Wise Analysis of Perturbation Robustness and Its Scaling

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Abstract

When a language model fails on surface-perturbed input (typos, OCR noise, homophones), “which layer is responsible” has three natural operationalizations: where representations diverge most (*sensitivity*), where restoring clean activations recovers the prediction (*causality*), and where a small adapter can repair the damage (*compensatory capacity*)—and we show these three layer maps dissociate. Across a five-model panel we identify two propagation regimes—*spike-and-suppress* (Phi-3.5, Gemma-2-9B) and *late-accumulation* (Llama-3, Mistral, Qwen2.5-7B)—and on the two models meeting an 80% identity-patch gate, sensitivity and causality are *anti-correlated* ($\rho = -0.72$ to -0.88). Within-family scaling on Qwen2.5 (1.5B→14B) shows the late-accumulation signature strengthening monotonically with scale, corroborated on a second family. We propose *cascade disruption* as the mechanism behind the dissociation: adapters placed at causally implicated early layers break intact downstream computation, making diagnostic-flagged sites the *worst* adapter placements. A fixed-harness layer sweep across four models (3.8–8B) confirms the core prediction on chain-of-thought GSM8K—the flagged sites are the most damaging adapter windows on every adjudicable model—and is sign-consistent but strongly attenuated on a multiple-choice control, consistent with damage that compounds with generation length. The sweep yields practical guidance: a training-free LRD pre-screen and a default-deepest placement rule, though absolute gains over no-adapter baselines remain small. Finally, apparent gains from a representation-stability loss reverse under an adequate generation budget—truncated chain-of-thought had been scored as empty—a methodological warning for any intervention evaluated on chain-of-thought tasks.

1 Introduction

A language model that solves a math problem will often fail on the same problem when the input contains a few typos. Surface perturbations—typos, OCR noise, speech-like substitutions, homophones—are among the most common corruptions models encounter in deployment (Pruthi et al., 2019; Rijhwani et al., 2020; Liu et al., 2023), and when they cause a failure, a natural question is *which layers are responsible*. Mechanistic analysis of transformer internals (Elhage et al., 2021) makes the question tractable, but “responsible” hides an ambiguity: it could mean the layers the perturbation moves most, the layers where restoring the clean representation recovers the correct output, or the layers where a lightweight adapter can compensate for the damage.

These three notions feel like they should agree: where the damage is largest, undoing it ought to help most, and a repair ought to be best spent. We measure all three per layer—**sensitivity**, via Layer-Wise Representation Divergence (LRD), a cosine divergence between clean and perturbed hidden states that builds on prior analyses of perturbation propagation (Moradi and Samwald, 2021; Shi and Huang, 2020); **causality**, via activation patching (Meng et al., 2022); and **compensatory capacity**, via per-window LoRA adapters—across a five-model panel. The maps do not agree (Figure 2). They peak at different layers, and on the models with the cleanest patching signal, sensitivity and causality are systematically *anti-correlated*.

The disagreement matters in practice. Interpretability diagnostics such as divergence probes and activation patching are increasingly used to decide *where to intervene*: where to place an adapter, where to apply representation engineering. Our results show that, for perturbation repair, these diagnostics point at the wrong layers. We propose a mechanism, *cascade disruption*: an adapter trained

at an early layer perturbs every layer downstream of it, breaking clean-input computation that was otherwise intact, so repair placed where the diagnostics point does net harm. A layer sweep under a fixed evaluation harness confirms this prediction on chain-of-thought GSM8K; on short-form MMLU the effect is sign-consistent but strongly attenuated. The flip side is actionable: a training-free LRD pre-screen rules out early- and mid-layer placement before any training, and the deepest window is consistently least regressive. The road to that sweep also produced a methodological warning: an intervention we ourselves once believed in—a representation-stability loss with apparently large gains—turned out to be an artifact of a generation budget too short for chain-of-thought.

Beyond the dissociation itself, the layer-wise view exposes structure. Models fall into two propagation regimes—*spike-and-suppress*, where early divergence is absorbed by mid-depth, and *late-accumulation*, where divergence compounds toward the output—and within a model family the late-accumulation signature strengthens monotonically with scale.

Our contributions are:

1. **Dissociation.** The three natural per-layer notions of “responsible for a perturbation failure”—sensitivity, causality, and compensatory capacity—produce distinct and, in key cases, anti-correlated layer maps. Evidence: both models passing the 80% identity-patch gate show strong sensitivity–causality anti-correlation (§4.4); convergent strands in §4.7.
2. **Regimes and scaling.** Models split into two propagation regimes, and the late-accumulation signature strengthens monotonically with scale within a family. Evidence: a three-point Qwen2.5 scan, monotone on every perturbation type, corroborated by a two-point Llama scan (§4.1.1).
3. **Mechanism, and a validated placement anti-pattern.** Cascade disruption—adapters at causally implicated early layers damage clean-input computation downstream—explains the dissociation and yields placement guidance: do *not* place adapters where divergence or patching point; default instead to the deepest window. Evidence: downstream disruption profiles (§4.5) and a fixed-harness layer sweep in which diagnostic-flagged sites are the most damaging windows (§5), width-

robust and worst of all when adaptation spans every layer.

4. **An evaluation protocol for chain-of-thought interventions (negative result).** Apparent intervention gains under a short-generation harness reverse under an adequate generation budget; the remedy is a generation-length-matched null. Evidence: the same checkpoint flips from +7.3 pp to −3.5 pp when the truncation artifact is fixed (§5.3).

2 Related Work

Robustness to surface perturbations. Character-level and lexical perturbations degrade NLP systems across tasks (Wang et al., 2022; Belinkov and Bisk, 2018; Eger et al., 2019; Li et al., 2020; Dong and Smith, 2018; Li et al., 2024). Rather than constructing worst-case inputs, we ask where in the network these failures localize.

Representational similarity. Probing (Belinkov, 2022), CKA (Kornblith et al., 2019), SVCCA (Raghu et al., 2017), and model stitching (Bansal et al., 2021) measure what models encode at each layer. LRD instead tracks *perturbation-induced* change at every layer, giving a propagation trajectory rather than a snapshot.

Layer-wise function in transformers. Transformer layers are functionally heterogeneous; early layers handle surface structure and later layers encode more abstract semantic information (Tenney et al., 2019; Jawahar et al., 2019; Geva et al., 2022, 2023; Dar et al., 2023). The three-map dissociation adds a perturbation-robustness dimension to this heterogeneity.

Activation patching and causal tracing. Replacing activations at specific layers with clean-run values is standard for causal localization (Vig et al., 2020; Meng et al., 2022; Wang et al., 2023; Hanna et al., 2023; Conmy et al., 2023). We show that causal localization does not transfer to repair: the layers patching implicates are the most damaging adapter sites in our sweep.

3 Method

3.1 Layer-Wise Representation Divergence (LRD)

For mean-pooled hidden states $\bar{\mathbf{h}}_{\text{clean}}^{(L)}$ and $\bar{\mathbf{h}}_{\text{noisy}}^{(L)}$ at layer L on a (clean, perturbed) input pair,

$$\text{LRD}(L) = 1 - \cos(\bar{\mathbf{h}}_{\text{clean}}^{(L)}, \bar{\mathbf{h}}_{\text{noisy}}^{(L)}). \quad (1)$$

Two summary statistics—*slope* (linear trend across layers) and *recovery rate* ($\text{LRD}(0) - \text{LRD}(L_{\max}) / \text{LRD}(0)$)—distinguish spike-and-suppress (high recovery, non-positive slope) from late-accumulation (near-zero recovery, positive slope); §4.1 applies this classification to the panel.

3.2 Perturbation Types

We use six perturbation types, five applied at character-level corruption rates—typos (5%), OCR (5%), whitespace (10%), case swap (10%), speech-like (10%)—and homophones applied at lexical (per-word) rates (20%/40%/50%). Unless noted, “Homophones” cells and the layer-sweep mean use the 20% severity.

3.3 Activation Patching

At each layer L we substitute the clean hidden state into the noisy forward pass and record the fraction of originally-failed examples (clean-correct, perturbed-incorrect) that now succeed. An identity (all-layer) patch gives the ceiling; a random-noise patch the floor. Layers with substantial single-layer recovery form the **causal window**. Because a low identity ceiling means the patching procedure itself is unreliable on that model, we gate results by it: identity recovery $\geq 80\%$ marks a model as *confirmed*, otherwise *exploratory*.

3.4 Adapter Setup and Three-Map Signals

We attach LoRA adapters (Hu et al., 2022) ($r=4$, $\alpha=8$), whose per-layer placement flexibility lets us measure compensatory capacity per layer, to attention $q_{\text{proj}}/v_{\text{proj}}$ (Phi-3.5 has fused attention, so we use $qkv_{\text{proj}}/o_{\text{proj}}$ equivalently) with the base model frozen, training on perturbed GSM8K with clean supervision targets and cross-entropy on answer positions. The principal sweep is CE-only; a representation-stability variant adding $L_{\text{stab}} = 1 - \cos(\bar{\mathbf{h}}^{\text{LoRA}}, \bar{\mathbf{h}}^{\text{frozen}})$ appears only for the ablation in §5.3. The three-map analysis compares three normalized per-layer signals: *sensitivity* ($\text{LRD}(L)$), *causality* (patching recovery at L), and *compensatory capacity* (perturbed-accuracy gain from a 5-layer LoRA window centered at L). Pairwise Spearman ρ is computed across layers per model; per-layer Cohen’s d (Cohen, 1988) at L is the standardized mean difference in $\text{LRD}(L)$ between failing and succeeding examples.

3.5 Cascade Disruption and Statistical Framework

Cascade disruption (§4.5) is the downstream LRD between frozen and adapted models on clean inputs. Two post-hoc intrinsic metrics probe whether frozen-model statistics predict adapter placement: **C3**, the entropy-based stable rank of the layer’s $q_{\text{proj}}/v_{\text{proj}}$ weights, and **C4**, the Frobenius norm of the teacher-forced cross-entropy gradient on the same weights (definitions, implementation details, and two further metrics, C1/C2, in Appendix F). C3/C4 correlated with LoRA effectiveness on the original three models but failed prospective validation (§4.4); we report them as descriptive only. Because layer-wise signals auto-correlate (partial-autocorrelation decay to the 95% white-noise band by lags 5–8), we use the **moving-block bootstrap** (Künsch, 1989) (block size 5, 10k resamples) for correlation CIs and p -values; post-decorrelation effective N is typically 3–5.

4 Diagnostic Results

Setup. Diagnostic runs use $n=500$ per condition ($n=200$ for the Llama-3 and Gemma-2-9B recovery profiles; Table 4) on Phi-3.5-mini-instruct (3.8B; Abdin et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Llama-3-8B-Instruct (Grattafiori et al., 2024), Gemma-2-9B base (Team et al., 2024), and Qwen2.5-7B-Instruct (Qwen et al., 2025) over GSM8K (Cobbe et al., 2021), MMLU (Hendrycks et al., 2021), and BBH (Suzgun et al., 2023). Clean GSM8K accuracies are in Table 1. A single fixed evaluation harness with sufficient `max_new_tokens` is used throughout (§5.3); full hyperparameters are in Appendix P (Table 11).

4.1 Two Propagation Regimes

LRD trajectories across the panel take exactly two shapes. In *spike-and-suppress* models, perturbation-induced divergence is large at the embedding and early layers and is then absorbed (a fifth to three fifths gone by the final layer). In *late-accumulation* models, early divergence is small but compounds with depth and never recovers. Table 1 classifies the five models by recovery rate and slope (§3). LRD heatmaps for all five models are in Appendix A (Figures 3–4).

Per-perturbation recovery rates confirm the classification (Appendix D, Table 4). The regime labels raise an immediate confound—within this panel, regime tracks model family—which the within-

Model	Regime	Clean	Recovery	Slope
Phi-3.5	S&S	85.4%	34–60%	≤ 0
Gemma-2-9B	S&S	64.8%	21–51%	≤ 0
Llama-3	Late	78.4%	0–5% [§]	+
Mistral	Late	59.6%	0–5% [§]	+
Qwen2.5-7B	Late	89.0%	0–5% [§]	+

Table 1: Five-model regime classification. S&S = spike-and-suppress. Recovery is the fraction of layer-0 divergence eliminated by the final layer (range across the six perturbation types); slope is the sign of the linear trend across layers. § Homophones (20% severity) are the one exception: recovery ≈ 35 –40% on all three late-accumulation models (Table 4); every other late-accumulation cell is near-zero.

family scaling scans below are designed to weaken.

4.1.1 Within-Family Scaling of the Late-Accumulation Regime

Holding family fixed and varying only scale deconfounds regime from architecture. The primary scan is Qwen2.5 at 1.5/7/14B (a $9\times$ range, all with ≥ 28 layers); a two-point Llama corroboration (1/8B) gives a second family with the same direction modulo a narrow-bin caveat at the 17-layer 1B point (Figure 1; per-perturbation grid in Appendix B, Table 3).

Qwen is treated as primary evidence and Llama as direction-only corroboration; neither is a controlled multi-family sweep, and we make no universal-scaling claim (Llama-3-8B uses $n=200$; its late/early ratio is stable to ~ 0.2 under bootstrap resampling). A Mann–Whitney check on final-layer LRD (wrong vs. right) for Llama-3.2-1B-Instruct ($n=500$) matches the Llama-3-8B per-perturbation pattern (4-of-6 significant).

4.2 Per-Layer Cohen’s d : Where the Failure Signal Emerges

If divergence causes failure, examples the perturbation breaks should be separable from examples that survive—and the depth at which they become separable marks where failures crystallize. We measure this with per-layer Cohen’s d : the standardized difference in LRD(L) between failing and succeeding examples (§3.4), shown per model in Appendix H (Figure 10). On Phi-3.5, d is near-zero through layers 0–14 for most perturbation types; succeed/fail separation develops at layers 20–27 and peaks at layers 28–32 ($d \approx 0.4$ – 0.6 ; homophones ≈ 0.9), downstream of the activation-patching causal window. The same deep-layer concentration appears

on the four other models. Two consequences follow: single-layer summaries are lossy (final-layer LRD alone predicts failure weakly), and the early layers patching will implicate (§4.3) lie *upstream* of where failures crystallize.

4.3 Activation Patching Locates the Causal Window

Single-layer clean patching (Appendix C, Figure 6) identifies the causal window on the two confirmed-gate models: Phi-3.5 ($n=50$ pairs, identity 100%) peaks at layers 1–8 and crosses baseline at $L18$; Llama-3-8B ($n=100$, identity 90%) peaks at $L1$ and stays above baseline through $L8$. Gemma-2-9B (identity 60%), Qwen2.5-7B (67%), and Mistral (identity recovery unstable across patching configurations, at most 65%) all fall below the 80% gate and are exploratory.

Qwen at scale clears the patching gate. On Qwen2.5-14B (48-layer sweep, 84 clean-correct pairs of 100), identity recovery rises to 85.0%, clearing the gate the 7B variant did not (67%). The full curve (Appendix C, Figure 7) is consistent with late-accumulation: early and mid thirds recover 73.1% and 71.1% of failed examples, while the late third falls to 42.3%. A higher-than-expected random-patch control (20.5%) weakens effect-size attribution but not the identity-ceiling promotion; the 14B curve is a within-family consistency check, not a third entry in the pre-registered two-model Spearman panel.

Note the tension already visible: patching implicates *early* layers, while the failure signal of §4.2 crystallizes *late*.

4.4 Three-Map Dissociation

When the three per-layer signals—LRD, patching recovery, LoRA effectiveness—are overlaid (Figure 2; all five models in Appendix G, Figure 9), their peaks do not coincide, and the relationships among them differ by regime.

LRD and patching are anti-correlated. On the two confirmed-gate models (Phi-3.5, Llama-3), $\rho = -0.72$ to -0.88 , $p < 0.001$ (eff $N \approx 3$ –5). Mistral matches the anti-correlation directionally but identity recovery falls below the 80% gate; we treat it as exploratory. Gemma-2-9B is an exception ($\rho = +0.389$); we conjecture its 60% identity ceiling corrupts the patching signal, and because Gemma fails the same gate, the exception sits out-

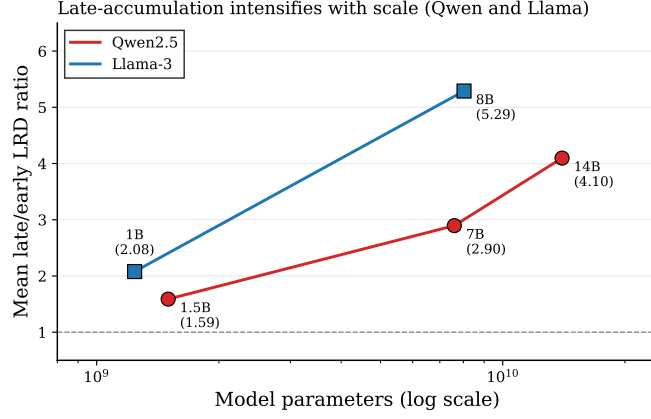
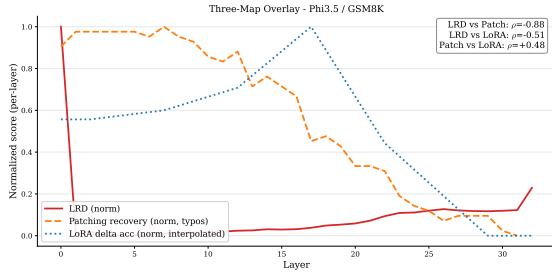
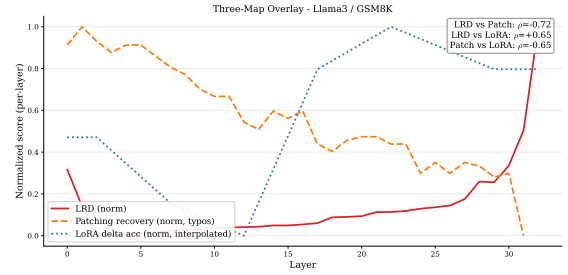


Figure 1: Within-family scaling of late accumulation. Mean late/early LRD ratio (deepest third over shallowest third, six-perturbation average) vs. parameter count (log x). Qwen2.5: 1.59 $\rightarrow$ 2.90 $\rightarrow$ 4.10; Llama: 2.08 $\rightarrow$ 5.29. Every per-perturbation cell is monotonic within family (Table 3).



(a) Phi-3.5 (spike-and-suppress): LoRA is most effective where LRD is suppressed.



(b) Llama-3 (late-accumulation): LoRA is most effective where LRD is still building.

Figure 2: **The three maps dissociate.** Per-layer sensitivity (LRD), causality (patching recovery), and compensatory capacity (LoRA effectiveness), one model per propagation regime; the signals peak at different layers, and sensitivity and causality are anti-correlated on both models. Remaining models: Figure 9; correlations: Table 5.

side the confirmed panel rather than contradicting it.

LRD $\times$ LoRA sign tracks regime. LoRA is most effective where LRD is suppressed (Phi-3.5, $\rho = -0.55$) or still accumulating (Llama-3/Mistral, $\rho = +0.63, +0.69$). After autocorrelation correction only Mistral reaches $p < 0.05$ individually ($p = 0.025$); we report the sign pattern across the panel rather than per-model significance. Mistral’s adapters here are from the same shared-grid learning rate later diagnosed as lr-sensitive (§5.2); we retain the correlation as directional evidence only, consistent with Mistral’s exclusion from formal sweep adjudication.

C3/C4 intrinsic metrics are post-hoc only. The frozen-model intrinsic metrics—weight effective rank (C3) and gradient norm (C4); §3.5—showed post-hoc correlation with LoRA effectiveness on the original three models ($|\rho| = 0.54\text{--}0.75$, regime-matched signs) but failed prospective validation:

C3/C4 predicted L08–11 for Qwen and L24–29 for Gemma, but the fixed-harness sweep (§5) finds Qwen L08–11 to be the *worst* window (-7.8 pp) and Gemma flat at every completed window. We treat C3/C4 as descriptive only (full correlations and the prediction-vs-outcome summary: Tables 5 and 7).

4.5 Cascade Disruption: Why Early Intervention Fails

If early-layer adapters fail because they break computation *downstream* of themselves, the damage should be directly measurable: an adapted model should diverge from the frozen model on *clean* inputs, at layers after the adapter window. For each trained LoRA window we compute the LRD between frozen and adapted models on clean inputs at every downstream layer (Appendix J, Figure 12).

The measurements confirm the mechanism’s signature: upstream of the adapted window, hidden states are bit-identical to the frozen model (diver-

gence exactly zero); disruption onsets at the window, persists through every downstream layer, and its total is largest for the earliest windows, shrinking as the window moves deeper on all four measured models (Phi-3.5, Qwen2.5-7B, Llama-3, Mistral; Appendix J, Figure 13). The rank order of total clean disruption across windows is inverse to that of perturbed-accuracy gain (Appendix K, Figure 14): placing LoRA at the layers patching implicates breaks clean-input processing, so the anti-correlation between the patching window and the best LoRA window is mechanistic, not coincidental. (All profiles use the independently retrained sweep adapters of Appendix K; both quantities come from the same checkpoints.)

Cascade disruption makes a falsifiable behavioral prediction: the windows the diagnostics flag should be the *most* damaging adapter sites, not the best. Before testing it (§5), we first check that the diagnostic maps themselves are not artifacts of the task or the perturbation distribution (§4.6).

4.6 Cross-Task and Held-Out Generalization

LRD profiles on Phi-3.5 and Mistral (the two models for which we collected cross-task data) are near-identical across GSM8K, MMLU, and BBH ($\rho > 0.95$ in both models): task modulates LRD magnitude but not profile shape (Appendix L, Figure 15). The regime signature is a property of the model, not of the task on which it is measured.

Nor is it a property of the perturbation distribution: we ran LRD on three held-out perturbations (char_insert, char_swap, qwerty; absent from training and the main analysis), one model per regime. Final-layer LRD on GSM8K: Phi-3.5 (S&S) 0.016, 0.023, 0.020; Llama-3-8B (Late) 0.066, 0.105, 0.082—a $\sim 5\times$ regime gap. Severity and error rate also dissociate: Llama-3-8B has higher LRD yet loses fewer of its clean-correct examples to the perturbation (87–90% survive, vs. 60–72% for Phi-3.5; each scored on the model’s own $n=200$ clean-correct subset), consistent with LRD measuring cascade severity rather than error rate.

4.7 How the Pieces Support the Dissociation Claim

The paper’s central claim is that sensitivity, causality, and compensatory capacity are dissociated per-layer properties. No single experiment above is decisive; the claim rests on six convergent strands, each ruling out a different deflationary explana-

tion. (i) *LRD $\times$ patching anti-correlation on both confirmed-gate models* (§4.4): if sensitivity and causality were one underlying map read through noise—noisy agreement—the correlation would be positive, not strongly negative. This comparison alone kills that alternative outright. (ii) *Regime-locked sign flip of LRD $\times$ LoRA* (§4.4): the LoRA map’s relationship to LRD flips sign with propagation regime—the dissociation has regime structure. (iii) *Cross-task profile stability* ($\rho > 0.95$ across GSM8K/MMLU/BBH; §4.6) rules out a single-task artifact. (iv) *Held-out perturbation types reproduce the regime gap* (§4.6), ruling out an artifact of the training perturbation distribution. (v) *Layer-sweep asymmetry* (§5): diagnostic-flagged sites are the *worst* adapter windows—a behavioral confirmation that reuses none of the correlational machinery. (vi) *Within-family scaling monotonicity* (§4.1.1) weakens the possibility that the taxonomy merely relabels architectural family.

The weak links, stated plainly: the effective sample size behind each correlation is only $\approx 3\text{--}5$ after decorrelation; the confirmed anti-correlation rests on the two gate-passing models (plus Mistral directionally); and the LRD $\times$ LoRA correlations mostly miss per-model significance—we claim the sign pattern, not the individual coefficients. The noisy-agreement alternative has to survive all six strands simultaneously—strand (i) alone already rules it out, and each remaining strand closes off a different deflationary explanation in turn.

5 Layer Sweep: Diagnostic-Flagged Sites Resist Repair

The cascade-disruption result (§4.5) predicts that the patching-implicated early layers should be *the worst*, not the best, sites for perturbation-correction adapters. We test it by sweeping adapter windows across the depth of Phi-3.5, Qwen2.5-7B, Mistral, and Llama-3 (3.8–8B) plus the 9B Gemma base, under the fixed harness used throughout. Two of the five are not adjudicable: Mistral (learning-rate artifact) and Gemma (adapter insensitivity; §5.2).

Setup. We train CE-only LoRA (configuration of §3.4; $\lambda_{\text{stab}}=0$) on perturbed GSM8K with clean supervision, sweeping non-overlapping 5-layer windows across the depth of each model with three seeds per window. The perturbed Δ is the mean change in perturbed accuracy over the six perturbation types relative to the no-adapter baseline, under the fixed harness (§5.3).

5.1 Per-Model Layer Sweeps

Table 2 gives the full grid; Figure 16 plots the sweep, and per-model panels with seed dispersion are in Figure 17 (both in Appendix M).

Per-model results. On Phi-3.5 (S&S; 32 layers) every mid-layer window is strongly negative; the only non-negative window is L27–31 (+0.2 pp, effectively null). Seed std (sample, $n-1$) ranges 0.1–1.1 pp, no window crosses zero under seed variation, and the earlier-harness L15–19 “optimum” falls to -3.5 under the fixed harness. Qwen2.5-7B’s worst window is the C3/C4-predicted L08–11 (-7.8 pp); its least-regressive is the deepest, L24–27 (+0.3). Llama-3-8B has L20–24 and L27–31 non-negative (+0.4, +1.1). Mistral-7B-v0.3 is uniformly negative at the shared lr; a single-cell audit at $lr=1e-5$ recovers to +0.13 pp, consistent with an lr artifact (§5.2). Gemma-2-9B is adapter-insensitive (every window within ± 0.7 pp of zero) and we treat it as “no measurable signal.”

5.2 The Mid-Layer Wall and the Pre-Registered Placement Tally

Two structural observations follow from Table 2. First, **mid-layer windows uniformly backfire on chain-of-thought GSM8K** (sign-consistent but strongly attenuated on short-form MMLU; see below): on all three adjudicable models, every window in the L05–L20 range is strongly negative, and the patching-implicated early windows (L00–04) are also negative, though less catastrophically. Second, **least-regressive windows are the deepest ones**: the only non-negative windows on the three models are the deepest available, confirming the cascade-disruption prediction that placement should be *away from* where divergence and patching causality concentrate.

Adjudicated tally. Excluding Mistral (lr artifact) and Gemma (flat at every window), the pre-registered prediction was that windows away from the patching-implicated early layers would be less damaging than the flagged layers. In addition, C3/C4—fit post hoc on the original three models—was registered prospectively for Qwen-7B before its sweep ran, marking L08–11 as its best window. The tally: Qwen ✓ (L24–27 beats L08–11 by 8.1 pp, and the C3/C4 pick is Qwen’s *worst* window); Llama-3 ✓ (L27–31 beats L05–09 by 7.5 pp); Phi-3.5 × (scored as a fail: no window rises meaningfully above zero, though

the relative ordering—mid worst, deepest least regressive—matches). Two holds and one fail. What the sweep delivers is a *validated placement anti-pattern with a mechanism*: do not place perturbation-repair adapters where LRD or patching point. It is a relative-ranking rule, not a positive recipe—absolute gains over the no-adapter baseline are small (Llama-3 +1.1 pp) or effectively null (Phi-3.5, Qwen), and Phi-3.5’s flatness is why its tally entry is a fail.

Width robustness and dose response. The wall is not an artifact of the fixed 5-layer window: a window-width ablation on Phi-3.5 (widths 3/5/7; Appendix N, Table 8) preserves the sign structure at every width, and mid-window damage grows monotonically with width ($-2.8 \rightarrow -5.3 \rightarrow -7.8$ pp). The all-layer baseline (Table 2, last column) extends this dose response to its endpoint: adapting every layer regresses perturbed accuracy more than any single window on all three adjudicable models, and regresses *clean* accuracy as well (-8.1 to -14.1 pp; Appendix N), as expected if the early-layer share of an all-layer adapter incurs cascade damage. Relative to all-layer adaptation, the deepest window is 9–14 pp less regressive while training a fraction of the parameters.

Beyond chain-of-thought: an MMLU control.

A fixed-harness sweep on MMLU (multiple-choice; one model per regime; Appendix N, Table 10) is sign-consistent with the GSM8K wall but strongly attenuated—we do not claim the wall replicates there. The only reliably nonzero cell is Phi-3.5’s diagnostic-flagged mid window L10–14 (-1.8 ± 0.4 pp); no window on either model produces positive transfer, so the absence of a positive placement recipe holds on both tasks. The attenuation is consistent with cascade disruption, which predicts damage compounds over autoregressive generation steps—long chain-of-thought generations should suffer far more than single-token multiple choice—an exploratory interpretation. MMLU’s short-answer format also makes generation-length harness artifacts (§5.3) structurally impossible, giving the sweep an artifact-immune task.

5.3 An Attempted Intervention and a Harness Artifact

We explored the representation-stability loss L_{stab} of §3.4 added to CE LoRA. An earlier (pre-fix) harness with `max_new_tokens=100` reported +7.3 pp on Phi-3.5 at L15–19 and a regime-dependent split.

Model (clean)	L00–04	L05–09	Early-mid	Mid	Mid-late	Late	All layers
Phi-3.5 (85.4%)	-1.4 ± 0.7	-0.9 ± 0.1	-5.3 ± 1.1	-3.5 ± 0.8	-0.1 ± 0.2	$+0.2 \pm 0.3$	-8.6 ± 0.7
Qwen2.5-7B (89.0%)	-0.6 ± 0.2	-4.0 ± 0.3	-7.8 ± 1.7	-3.4 ± 0.2	-0.3 ± 0.1	$+0.3 \pm 0.0$	-13.6 ± 0.1
Llama-3-8B (78.4%)	-2.5 ± 0.2	-6.4 ± 0.5	—	-2.7 ± 0.8	$+0.4 \pm 0.2$	$+1.1 \pm 0.1$	-9.9 ± 1.0
Mistral-7B-v0.3 (59.6%)	-7.8 ± 2.5	-17.7 ± 1.1	—	-19.4 ± 0.9	-8.5 ± 0.4	-4.3 ± 0.3	—
Gemma-2-9B (64.8%) ^s	+0.2	-0.2	—	+0.5	+0.2	0.0	—

Table 2: Mean perturbed Δ (with-adapter minus no-adapter, six perturbations on 500 GSM8K items); mean $\pm$ sample std over three seeds (^s: Gemma single-seed). Headers are nominal; exact windows, the Gemma protocol, and the pre-registered Qwen L08–11 cell (C3/C4’s pick—Qwen’s worst window) are in Appendix M. Dashes: not trained. *Mistral excluded from adjudication (lr artifact; §5.2)*. “All layers” adapts every layer; it regresses more than any single window on all three adjudicable models.

Under the fixed harness those gains reverse (Appendix O, Figure 18; max_new_tokens=512 for this checkpoint comparison and the layer sweep of §5; settings in Appendix P): Phi-3.5 L15–19 records -3.5 pp, and the Qwen2.5-7B L24–27 vanilla CE baseline reads -7.5 pp where the same checkpoint had read $+11.6$ pp. (The -7.5 re-scores the *earlier pipeline*’s checkpoint; Table 2’s $+0.3$ for the same window comes from adapters trained afresh.) The cause is mundane: truncated chain-of-thought was scored as empty, crediting conditions that produce shorter (often less accurate) output.

The reversal is not merely bookkeeping. Holding the fixed harness constant and toggling only λ_{stab} , the stability term ($\lambda_{\text{stab}} = 1.0$) *hurts* perturbed accuracy by 1.2–3.9 pp across seeds on Phi-3.5 L15–19—the window where the earlier harness reported $+7.3$ pp. The diagnostic, mechanism, and scaling results in §4 are independent of this claim; the reversal rules out a regime-dependent stabilizer benefit, leaving cascade disruption—which predicts mid-layer harm—as the surviving explanation.

6 Discussion

Restoration vs. adaptation. Why should the causal map mislead? Patching substitutes the *exact* clean representation at inference time—an oracle intervention—so it identifies where *restoration* helps. An adapter must instead *learn* a correction applied to every input, including clean ones, with low-rank weights. Early layers are where restoration is cheapest but also where a learned modification propagates through the most downstream computation; the layer sweep shows that cost dominates, damage scaling monotonically with adapted-window width (§5.2).

Implications for practitioners. First, LRD is a cheap pre-screen: forward passes on

clean/perturbed pairs identify a model’s propagation regime—and rule out early- and mid-layer adapter placement—before any training compute is spent. Second, diagnostic-guided placement is falsified in this setting: neither activation patching nor intrinsic weight metrics (C3/C4) select good adapter sites, and both actively select bad ones; absent task-specific evidence, default to the deepest window (a relative ranking; absolute gains remain small or null). Third, any intervention evaluated on chain-of-thought tasks should ship with a generation-length-matched null; re-scoring checkpoints under an adequate budget is nearly free.

Open questions. Regime initially tracked family within our panel; the within-family scans weaken that confound but do not eliminate it, and we make no universal-scaling claim. Two follow-ups: a controlled multi-family scan varying architecture and scale independently, and a sweep across tasks of intermediate generation length to test the compounding-damage account directly.

7 Conclusion

Which layer is “responsible” for a perturbation failure depends on what responsible means: sensitivity, causality, and compensatory capacity give three different per-layer answers, and the disagreement is systematic. The regimes are stable across tasks and perturbation types and strengthen with scale within two families. Cascade disruption explains why causally implicated layers resist repair; a fixed-harness sweep confirms its central prediction—diagnostic-flagged sites are the most damaging adapter windows—on every adjudicable model—and an intervention we once believed in did not survive re-evaluation under an adequate generation budget.

Limitations

Scope of intervention claims. We provide no positive intervention recipe. All placement-related claims are relative-ranking statements under a fixed harness (“window A is less regressive than window B ”); absolute gains over no-adaptor baselines are not claimed. The earlier-pipeline representation-stability gains were a harness artifact (§5.3); the underlying cascade-disruption mechanism is independently supported by the clean-input disruption measurements (§4.5).

Panel size. Our regime taxonomy is observed on a five-model panel. The primary de-confounding evidence is the three-point Qwen2.5 scan (§4.1.1); the two-point Llama scan corroborates the direction on a second family but has a narrow-bin caveat at the 1B end (17 hidden layers). Two families is not a controlled multi-family sweep, and confirming regime as a general transformer property still requires within-family scans on additional families and, ideally, controlled pretraining recipes.

Task and generation-length scope. The layer-sweep adjudication uses GSM8K, with MMLU as a short-form control on one model per regime (§5.2). Mid-layer damage magnitude appears generation-length dependent: on MMLU the only reliably nonzero window is Phi-3.5’s mid window (−1.8 pp) and all other windows on both models are within noise, so effects on short-form and multiple-choice tasks are strongly attenuated—tested on MMLU only, with no sweep at intermediate generation lengths. Cross-task LRD stability is measured on two models (Phi-3.5 and Mistral; §4.6) across GSM8K/MMLU/BBH. All experiments are English-only.

Patching ceiling effects. Gemma-2-9B and Qwen2.5-7B identity ceilings (60%, 67%) limit causal interpretation when perturbation effects distribute across layers. Mistral’s identity recovery varies across patching configurations (at most 65%, with large run-to-run variance), so we report no single ceiling for it. Qwen-14B clears the 80% gate (85.0%) but with a 20.5% random-patch warning that weakens effect-size attribution on the 14B curve.

Statistical power. Three seeds per layer-sweep window characterizes each cell’s direction and magnitude but limits the precision of fine-grained between-window comparisons. Activation-

patching estimates likewise rest on small pair counts (Phi-3.5 $n=50$, Llama-3 $n=100$, Qwen-14B 84 pairs). The main sweep uses a fixed 5-layer window; the width ablation (§5.2, Table 8) shows the sign structure is width-robust at 3/5/7, but covers one model (Phi-3.5). The block-bootstrap correlation analyses use block size 5 from the partial-autocorrelation decay; the post-correction effective $N \approx 3-5$ is already small. A block-size sensitivity sweep ($b=2-8$; Appendix 8) shows the primary LRD-vs-patching anti-correlation stays significant ($p < 0.05$) at every b for all three models, while the LRD-vs-LoRA and patching-vs-LoRA pairs are not uniformly significant across b —consistent with the sign-pattern-not-coefficients framing above.

Mistral learning-rate sensitivity. At the shared-grid lr ($5e-5$) every Mistral window is strongly negative; a single-cell re-run at $1e-5$ recovers to +0.13 pp, consistent with an lr artifact rather than a capacity issue, though we did not re-run every window. Mistral is excluded from the placement-prediction adjudication.

Code and data. Code, configurations, and per-model evaluation logs are available at <https://github.com/NathanLabiosa/Layer-Wise-Representation-Divergence>.

Use of AI assistants. AI coding assistants were used during this project for code scaffolding, experiment orchestration (job-launch scripts, log parsing, plot generation), and writing assistance (copy-editing and reorganization of paper text). All scientific claims, experimental designs, statistical analyses, and result interpretations are the authors’ own; AI-generated text and code were reviewed and verified by the authors before inclusion.

Artifact use and licensing. All models and benchmarks used here are publicly released research artifacts, cited at first use in §4 and listed in references. Licenses, to the best of our knowledge at the time of writing: Phi-3.5-mini-instruct (MIT), Mistral-7B-Instruct-v0.3 (Apache 2.0), Llama-3-8B-Instruct and Llama-3.2-1B-Instruct (Llama-3 Community Licenses), Gemma-2-9B (Gemma Terms of Use), Qwen2.5-{1.5B, 7B, 14B}-Instruct (Apache 2.0), GSM8K (MIT), MMLU (MIT), BBH (Apache 2.0). Our usage—measuring representational properties and training diagnostic LoRA adapters for academic research—is consistent with the intended use stated by each artifact’s release.

The benchmarks (GSM8K, MMLU, BBH) consist of grade-school math word problems, multiple-choice exam questions, and reasoning challenges respectively, and to our knowledge contain no personally identifying information or offensive content; each is documented in its release paper (cited).

Potential risks. This work is diagnostic and does not release new models, training data, or adversarial tools. The perturbation types we study (typos, OCR noise, whitespace, case, speech-like substitutions, homophones) are commonplace and not novel attack vectors. The diagnostic itself (LRD, patching recovery, layer sweeps) could in principle inform adversarial perturbation design or targeted intervention against deployed models; we judge this risk low because the techniques and perturbations are already public. The more practical risk is *misuse of the regime taxonomy*: extrapolating the spike-and-suppress / late-accumulation labels to models outside our five-model panel without re-running the diagnostic, or treating the post-hoc C3/C4 metrics as a placement rule despite the falsification in §4.4. The Limitations above are written to make both kinds of over-reach explicit. Finally, the layer-sweep and diagnostic experiments are GPU-intensive (multi-model LoRA sweeps with three seeds per window, plus activation patching at every layer); we report exact configurations in Appendix P so practitioners can scope re-runs to the cells they need.

References

- Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, Alon Benhaim, Misha Bilenko, Johan Bjorck, Sébastien Bubeck, Martin Cai, Qin Cai, Vishrav Chaudhary, Dong Chen, Dongdong Chen, and 110 others. 2024. [Phi-3 technical report: A highly capable language model locally on your phone](#). *Preprint*, arXiv:2404.14219.
- Yamini Bansal, Preetum Nakkiran, and Boaz Barak. 2021. [Revisiting model stitching to compare neural representations](#). In *Advances in Neural Information Processing Systems*, volume 34. Curran Associates, Inc.
- Yonatan Belinkov. 2022. [Probing classifiers: Promises, shortcomings, and advances](#). *Computational Linguistics*, 48(1):207–219.
- Yonatan Belinkov and Yonatan Bisk. 2018. [Synthetic and natural noise both break neural machine translation](#). In *International Conference on Learning Representations*.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training verifiers to solve math word problems](#). *Preprint*, arXiv:2110.14168.
- Jacob Cohen. 1988. *Statistical Power Analysis for the Behavioral Sciences*, 2nd edition. Lawrence Erlbaum Associates, Hillsdale, NJ.
- Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. 2023. [Towards automated circuit discovery for mechanistic interpretability](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Guy Dar, Mor Geva, Ankit Gupta, and Jonathan Berant. 2023. [Analyzing transformers in embedding space](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16124–16170, Toronto, Canada. Association for Computational Linguistics.
- Rui Dong and David Smith. 2018. [Multi-input attention for unsupervised OCR correction](#). In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2363–2372, Melbourne, Australia. Association for Computational Linguistics.
- Steffen Eger, Gözde Gül Şahin, Andreas Rücklé, Ji-Ung Lee, Claudia Schulz, Mohsen Mesgar, Krishnkant Swarnkar, Edwin Simpson, and Iryna Gurevych. 2019. [Text processing like humans do: Visually attacking and shielding NLP systems](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 1634–1647, Minneapolis, Minnesota. Association for Computational Linguistics.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, and 6 others. 2021. [A mathematical framework for transformer circuits](#). *Transformer Circuits Thread*. <https://transformer-circuits.pub/2021/framework/index.html>.
- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. 2023. [Dissecting recall of factual associations in auto-regressive language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore. Association for Computational Linguistics.

- Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. 2022. [Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 30–45, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.
- Michael Hanna, Ollie Liu, and Alexandre Variengien. 2023. [How does GPT-2 compute greater-than?: Interpreting mathematical abilities in a pre-trained language model](#). In *Thirty-seventh Conference on Neural Information Processing Systems*.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. [Measuring massive multitask language understanding](#). In *International Conference on Learning Representations*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.
- Ganesh Jawahar, Benoît Sagot, and Djamé Seddah. 2019. [What does BERT learn about the structure of language?](#) In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3651–3657, Florence, Italy. Association for Computational Linguistics.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. [Mistral 7b](#). *Preprint*, arXiv:2310.06825.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. [Similarity of neural network representations revisited](#). In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR.
- Hans R. K  unsch. 1989. [The jackknife and the bootstrap for general stationary observations](#). *The Annals of Statistics*, 17(3):1217–1241.
- Changye Li, Weizhe Xu, Trevor Cohen, and Serguei Pakhomov. 2024. [Useful blunders: Can automated speech recognition errors improve downstream dementia classification?](#) *Journal of Biomedical Informatics*, 150:104598.
- Linyang Li, Ruotian Ma, Qipeng Guo, Xiangyang Xue, and Xipeng Qiu. 2020. [BERT-ATTACK: Adversarial attack against BERT using BERT](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6193–6202, Online. Association for Computational Linguistics.
- Hanyu Liu, Chengyuan Cai, and Yanjun Qi. 2023. [Expanding scope: Adapting English adversarial attacks to Chinese](#). In *Proceedings of the 3rd Workshop on Trustworthy Natural Language Processing (TrustNLP 2023)*, pages 276–286, Toronto, Canada. Association for Computational Linguistics.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2022. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35.
- Milad Moradi and Matthias Samwald. 2021. [Evaluating the robustness of neural language models to input perturbations](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1558–1570, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Danish Pruthi, Bhuwan Dhingra, and Zachary C. Lipton. 2019. [Combating adversarial misspellings with robust word recognition](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5582–5591, Florence, Italy. Association for Computational Linguistics.
- Qwen, :, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, and 25 others. 2025. [Qwen2.5 technical report](#). *Preprint*, arXiv:2412.15115.
- Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. 2017. [Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability](#). In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Shruti Rijhwani, Antonios Anastasopoulos, and Graham Neubig. 2020. [OCR Post Correction for Endangered Language Texts](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5931–5942, Online. Association for Computational Linguistics.
- Zhouxing Shi and Minlie Huang. 2020. [Robustness to modification with shared words in paraphrase identification](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 164–171, Online. Association for Computational Linguistics.
- Mirac Suzgun, Nathan Scales, Nathanael Sch  rli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi,

Denny Zhou, and Jason Wei. 2023. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13003–13051. Association for Computational Linguistics.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, and 179 others. 2024. [Gemma 2: Improving open language models at a practical size](#). *Preprint*, arXiv:2408.00118.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. 2019. [BERT rediscovers the classical NLP pipeline](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, Florence, Italy. Association for Computational Linguistics.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. 2020. [Investigating gender bias in language models using causal mediation analysis](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc.

Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023. [Interpretability in the wild: a circuit for indirect object identification in GPT-2 small](#). In *International Conference on Learning Representations*.

Xuezhi Wang, Haohan Wang, and Diyi Yang. 2022. [Measure and improve robustness in NLP models: A survey](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4569–4586, Seattle, United States. Association for Computational Linguistics.

This appendix provides extended results referenced in the main paper. §A contains LRD heatmaps for all five panel models, and §B adds Qwen-1.5B and Qwen-14B heatmaps that complement the within-family scaling analysis (Figure 1, §4.1.1). §C gives the full 48-layer Qwen2.5-14B patching curve summarized in §4.3. §D–§F give per-perturbation recovery rates, the full panel of pairwise Spearman correlations across all five models, and the intrinsic-predictor (C1–C4) definitions and C3/C4 falsification table. §G–§J contain per-model figure panels for the three-map overlay (all five models), Cohen’s d (all five models), intrinsic capacity metrics (Llama-3/Mistral), and cascade disruption (Llama-3/Mistral). §K–§L provide the disruption-vs-gain scatter (Phi-3.5, Llama-3, Mistral) and the cross-task LRD profiles (Phi-3.5, Mistral). §M contains the per-model layer-sweep panels underlying Table 2. §O briefly catalogs investigations performed before the evaluation harness was fixed—the cosine vs. MSE stability-loss matched-rate controls, the λ_{stab} scaling sanity-check, the LoRA capacity experiments, and a residual-adaptor stabilizer tested on Mistral and Qwen2.5-7B—and explains why those numbers are excluded from this revision. §P lists reproducibility settings for the fixed-harness runs.

A LRD Profiles: All Five Panel Models

Figure 3 shows LRD heatmaps for the two models that anchor the regime taxonomy (§4.1): Phi-3.5 (spike-and-suppress) and Llama-3 (late-accumulation). Figure 4 shows the remaining three models. Gemma-2-9B exhibits the spike-and-suppress profile (large early divergence, suppressed through the middle layers, with a mild secondary hump near the final layers). Qwen2.5-7B exhibits the late-accumulation profile, with an early transient spike at layers 0–3 and dominant late buildup through layers 20–28. Mistral shows a gradual late-accumulation buildup through depth.

B Within-Family Scaling: Qwen-1.5B and Qwen-14B Heatmaps

Figure 5 shows LRD heatmaps for Qwen-1.5B and Qwen-14B; the Qwen-7B heatmap is in Figure 4b. Together with the within-family scaling figure in the main paper (Figure 1), these visualize the same late-accumulation regime at three Qwen scales: an early transient spike near layers 0–3 that fades, followed by a depth-dominant build-

up whose late/early ratio intensifies monotonically with scale ($1.59 \rightarrow 2.90 \rightarrow 4.10$). Table 3 gives the per-perturbation grid behind the main-paper scaling figure (Figure 1, §4.1.1).

C Activation-Patching Recovery Curves

Figure 6 shows the single-layer patching recovery curves for the two confirmed-gate models (§4.3).

Figure 7 gives the full 48-layer activation-patching recovery curve for Qwen2.5-14B on GSM8K typos 5% (84 clean-correct pairs of 100 attempted). The main paper (§4.3) summarises the curve into per-region recovery and the per-layer peak ($L23$, 82.1%) and worst ($L47$, 26.2%); the figure shows that the curve is broadly consistent with late-accumulation, with the highest recovery concentrated in the early-to-mid third and a clear collapse in the deepest layers.

D Per-Perturbation Recovery Rates

Table 4 gives per-perturbation recovery rates for all five models (GSM8K). Cascade slope summaries are in Table 1.

E Pairwise Spearman Correlations: All Five Models

Table 5 reports the full pairwise correlation panel underlying §4.4: for each model, Spearman ρ across layers between each pair of the three signals, with CIs and p -values from the moving-block bootstrap (§3.5). The pattern the main text summarizes is visible directly: $\text{LRD} \times \text{Patch}$ is strongly negative on the confirmed-gate models (Phi-3.5, Llama-3) and directionally negative on Mistral; $\text{LRD} \times \text{LoRA}$ flips sign with regime but reaches per-model significance only on Mistral (adapters from the same shared-grid later diagnosed as lr-sensitive, §5.2; retained here as directional evidence, consistent with Mistral’s exclusion from sweep adjudication); and the two models that fail the identity-patch gate (Gemma-2-9B, Qwen2.5-7B) produce weak or unstable correlations, consistent with a corrupted patching signal rather than a contradiction.

F Intrinsic Predictor Details

We evaluated four intrinsic metrics as candidate predictors of per-layer LoRA effectiveness. C3 and C4 are defined in §3.5 of the main paper; we repeat them here with implementation detail, alongside C1 and C2.

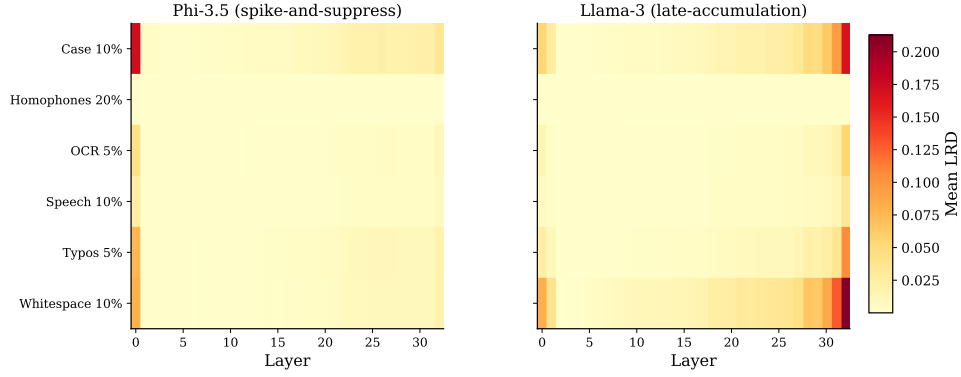


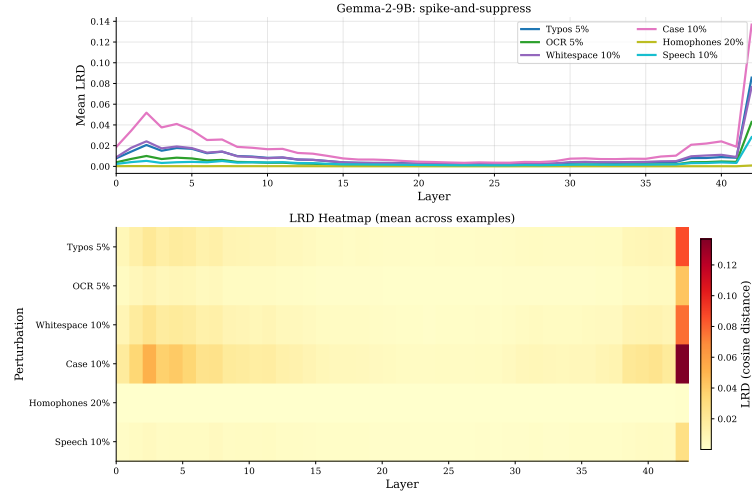
Figure 3: LRD heatmaps (GSM8K, typos 5%). Phi-3.5 shows spike-and-suppress; Llama-3 shows late-accumulation. Color encodes LRD magnitude on a shared scale; rows are layers, columns are examples.

Family	Model	Typos	OCR	Whitespace	Case	Homophones	Speech	Mean
Qwen2.5	1.5B	1.65	1.53	1.68	1.42	1.34	1.92	1.59
	7B	3.19	3.40	2.66	2.40	2.24	3.49	2.90
	14B	4.20	4.88	3.55	3.17	3.84	4.93	4.10
Llama	3.2-1B	2.44	2.02	2.24	2.51	1.28	1.98	2.08
	3-8B [§]	5.90	5.86	4.50	5.24	3.25	6.99	5.29

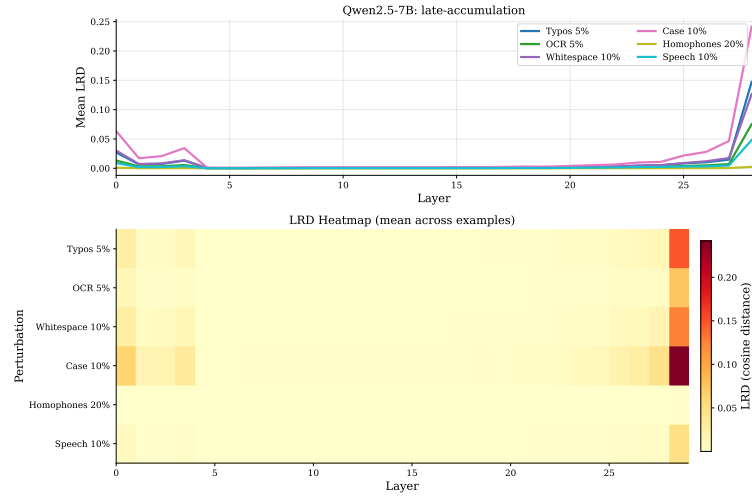
Table 3: Late/early LRD ratio per perturbation for the two scaling scans; Mean averages the six entries. Every per-perturbation cell is monotonic within family. GSM8K, $n=500$, $\text{max_new_tokens}=768$; [§] Llama-3-8B used $n=200$.

Perturbation	Phi-3.5	Gemma-2-9B	Llama-3	Mistral	Qwen2.5-7B
Typos 5%	39.5%	34.5%	0.5%	1.5%	1.8%
OCR 5%	60.0%	35.5%	3.5%	4.5%	5.2%
Whitespace 10%	43.5%	38.5%	0.0%	0.0%	0.8%
Case 10%	47.5%	36.5%	0.0%	0.0%	0.0%
Homophones 20%	56.5%	51.5%	34.5%	38.0%	39.8%
Speech 10%	34.5%	21.0%	1.0%	0.5%	1.6%

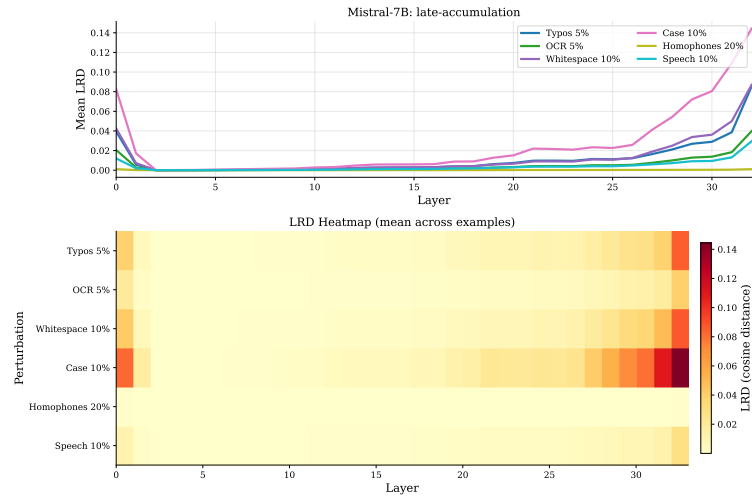
Table 4: Per-perturbation recovery rates for all five models (GSM8K). Recovery is the fraction of layer-0 divergence eliminated by the final layer. Spike-and-suppress models (Phi-3.5, Gemma-2-9B) show partial recovery on all types; late-accumulation models (Llama-3, Mistral, Qwen2.5-7B) show near-zero recovery, except homophones at 20% severity (the only condition producing meaningful late-model recovery). Sample sizes: Phi-3.5, Mistral, Qwen2.5-7B $n = 500$; Llama-3, Gemma-2-9B $n = 200$.



(a) Gemma-2-9B. Spike-and-suppress profile with a mild secondary hump at layers 30–41 and a sharp jump at the final layer.

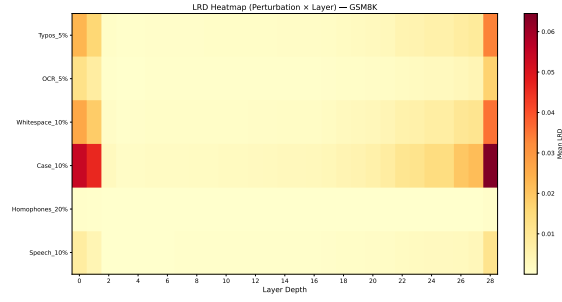


(b) Qwen2.5-7B. Late-accumulation profile with an early transient spike at layers 0–3 and dominant late buildup through layers 20–28.

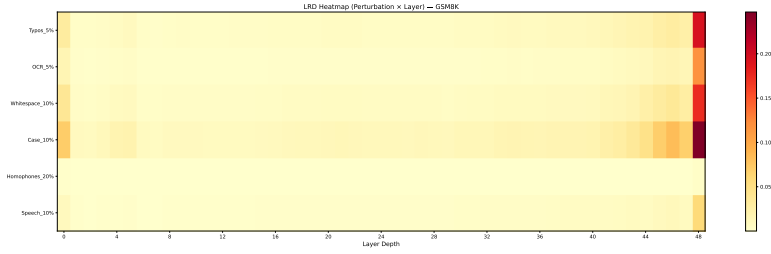


(c) Mistral. Late-accumulation profile; gradual buildup through depth.

Figure 4: LRD heatmaps (GSM8K, typos 5%) for the three additional models. Compare Phi-3.5 and Llama-3 in Figure 3.



(a) Qwen2.5-1.5B (late/early ratio 1.59). Late-accumulation profile is present but mild; depth-driven divergence is visible after roughly the midpoint of layers.



(b) Qwen2.5-14B (late/early ratio 4.10). Late-accumulation profile is sharper: deep-layer divergence dominates earlier and on every perturbation type.

Figure 5: LRD heatmaps (GSM8K, typos 5%) for Qwen2.5-1.5B and Qwen2.5-14B. Compare with Qwen2.5-7B in Figure 4b. The late-accumulation regime persists at all three scales and intensifies monotonically with size.

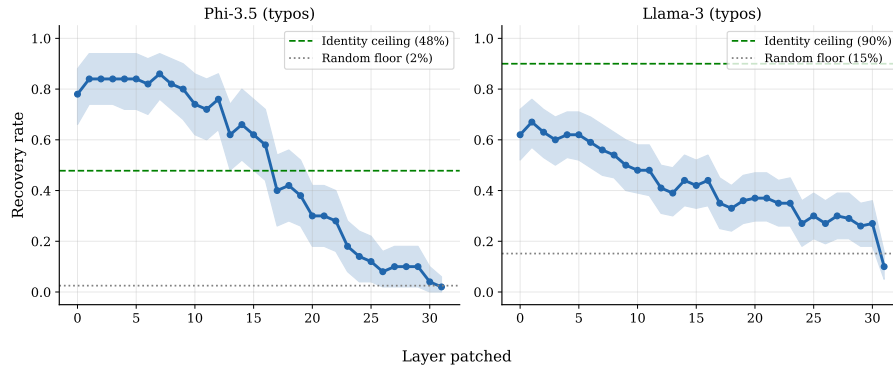


Figure 6: Activation patching recovery on the two confirmed-gate models (Phi-3.5, Llama-3; 3.8–8B). y -axes differ by identity ceiling (Phi-3.5 100%, Llama-3 90%). Early-layer patches recover the most failed examples.

Model	Pair	ρ	95% CI	p
Phi-3.5	LRD $\times$ Patch	-0.881	$[-0.991, -0.787]$	$<0.001^{***}$
	LRD $\times$ LoRA	-0.550	$[-0.884, +0.624]$	0.33
	Patch $\times$ LoRA	+0.482	$[-0.840, +0.909]$	0.44
Llama-3	LRD $\times$ Patch	-0.723	$[-0.982, -0.450]$	$<0.001^{***}$
	LRD $\times$ LoRA	+0.631	$[-0.018, +0.908]$	0.054
	Patch $\times$ LoRA	-0.642	$[-0.760, +0.091]$	0.072
Mistral	LRD $\times$ Patch	-0.752	$[-0.969, -0.550]$	$<0.001^{***}$
	LRD $\times$ LoRA	+0.685	$[+0.132, +0.854]$	0.025*
	Patch $\times$ LoRA	-0.731	$[-0.824, -0.080]$	0.033*
Gemma-2-9B	LRD $\times$ Patch	+0.389	$[-0.43, +0.83]$	0.264
	LRD $\times$ LoRA	+0.091	$[-0.38, +0.69]$	0.418
	Patch $\times$ LoRA	+0.735	$[+0.26, +0.92]$	0.006**†
Qwen2.5-7B	LRD $\times$ Patch	+0.025	$[-0.575, +0.728]$	0.866
	LRD $\times$ LoRA	+0.413	$[-0.763, +0.726]$	0.802
	Patch $\times$ LoRA	-0.167	$[-0.529, +0.356]$	0.602

Table 5: Pairwise Spearman correlations (block bootstrap, block size 5, 10k resamples). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. † Gemma-2-9B patching is exploratory (60% identity ceiling). LRD $\times$ LoRA is directionally consistent with regime but significant only on Mistral after autocorrelation correction; Mistral’s LoRA data here is from the same shared-grid lr later diagnosed as lr-sensitive (§5.2) and is retained as directional/exploratory evidence only.

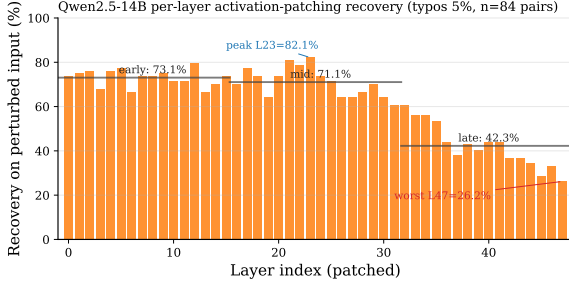


Figure 7: Qwen2.5-14B activation-patching recovery across all 48 layers on GSM8K typos 5%. Identity recovery 85.0% (above the 80% gate). The peak is at L23 (82.1%); the worst single layer is L47 (26.2%). The random-patch control on this model recovered 20.5% (logged warning; §4.3).

- **C1: Update ratio.** $C_1(L) = \|\mathbf{h}^{(L)} - \mathbf{h}^{(L-1)}\|_2 / \|\mathbf{h}^{(L-1)}\|_2$, where $\mathbf{h}^{(L)}$ is the hidden state at layer L , mean-pooled over valid (non-padded) token positions per example. Computed on $n = 200$ clean GSM8K examples and averaged across examples.
- **C2: Attention entropy.** $C_2(L) = -\sum_k p_k \log p_k$, where p_k are the attention probabilities at layer L (extracted via output_attentions=True). Entropy is computed per-head per-query-position, then averaged across (batch, heads, query positions) on $n = 200$ clean GSM8K examples.
- **C3: Weight effective rank (stable rank).** $C_3(L) = \exp(-\sum_i \tilde{\sigma}_i \log \tilde{\sigma}_i)$, with $\tilde{\sigma}_i =$

$\sigma_i / \sum_j \sigma_j$ and σ_i the singular values of the layer’s weight matrix. This is a smooth, differentiable rank proxy (not a threshold-based count). Computed statically on q_proj and v_proj; for Phi-3.5 fused qkv_proj, the Q and V slices are extracted and the per-layer score is $(r_Q + r_V)/2$. No forward pass required.

- **C4: Gradient norm.** $C_4(L) = \frac{1}{2} (\|\partial \mathcal{L}_{CE} / \partial W_Q^{(L)}\|_F + \|\partial \mathcal{L}_{CE} / \partial W_V^{(L)}\|_F)$, the mean of the Frobenius gradient norms of the Q and V weight matrices under teacher-forced cross-entropy loss. Computed on $n = 50$ clean GSM8K examples (subset of the 200) with batch size 2; requires forward+backward pass.

C3 and C4 show the strongest *post-hoc* correlations with LoRA effectiveness on the original three models (Table 6); C1 and C2 are reported in the same table for completeness. As reported in the main paper (§4.4), the prospective validation on Qwen-7B (C3/C4 predicted L08–11) is falsified by the fixed-harness layer sweep: L08–11 is the worst window for Qwen-7B (−7.8 pp mean perturbed Δ , three seeds), not the best. We do not recommend C3/C4 as a placement rule.

Block bootstrap sensitivity — block size 2–8, $n_{\text{boot}}=10,000$

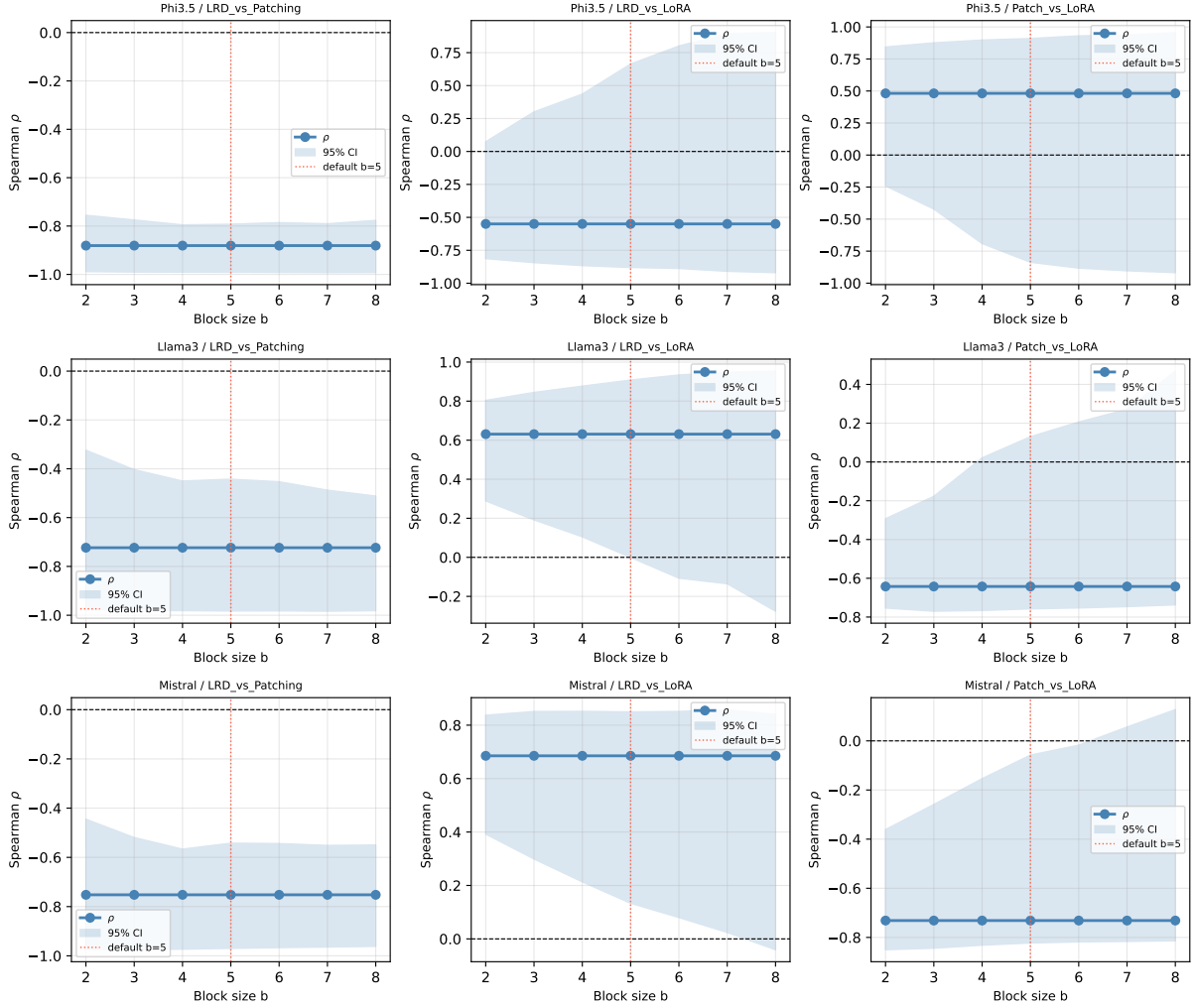


Figure 8: Block-size sensitivity for the moving-block bootstrap (Table 5), $b=2-8$, 10k resamples per b . Point estimate ρ is a property of the fitted layer-pair (not the resample), so it is constant across b ; the shaded band is the 95% CI. The LRD $\times$ Patch anti-correlation clears $p < 0.05$ at every b for all three models (dashed red line marks the reported default $b=5$); LRD $\times$ LoRA and Patch $\times$ LoRA lose significance at some block sizes, matching the sign-not-magnitude claim in §3.5.

Metric	Phi-3.5 ρ	Llama-3 ρ	Mistral ρ
C1: Update ratio	−0.517**	+0.394*	+0.430*
C2: Attn entropy	+0.211	−0.712***	−0.296
C3: Weight eff. rank	−0.627***	+0.751***	+0.670***
C4: Gradient norm	+0.537**	−0.714***	−0.594***

Table 6: Intrinsic capacity metric correlations with LoRA effectiveness (original three models; i.i.d. bootstrap, anti-conservative given $N \approx 3-5$; significance markers are indicative). C3 and C4 flip sign between Phi-3.5 and the late-accumulation models.

Table 7 summarises C3/C4 predictions versus the least-regressive window under the fixed-harness layer sweep (§5). The original-three-model fits stay post-hoc; on the two pre-registered models, the prediction is falsified on Qwen-7B and is moot

on Gemma (flat sweep).

Model	C3/C4 predicted	Least-regr. window	Outcome
Phi-3.5	L15–19	L27–31	Post-hoc only; sign flipped
Llama-3	L20–24	L27–31	Post-hoc only; close, not best
Mistral	L20–24	L27–31	Post-hoc only; close, not best
Qwen2.5-7B	L08–11	L24–27	Falsified (predicted = worst)
Gemma-2-9B	L24–29	(flat)	Moot ($\rho \approx 0$)

Table 7: C3/C4 predicted window versus the fixed-harness least-regressive window. The Phi-3.5/Llama-3/Mistral entries are post-hoc fits to the broken-harness sweep; under the fixed harness no window is non-negative on Phi-3.5 or Mistral, and the least-regressive window is L27–31 on all three. On Qwen-7B, the C3/C4 prediction is the *worst* window in the sweep (−7.8 pp; pre-registered falsifier). Gemma’s sweep is flat.

G Three-Map Overlays: All Five Models

Figure 9 shows the per-layer overlay of the three signals (LRD sensitivity, patching recovery, LoRA effectiveness) for each of the five models in the panel; the Phi-3.5 and Llama-3 panels also appear as the anchor figure in the main paper (Figure 2). Numerical correlations across the panels are in Table 5.

H Per-Layer Cohen’s d : All Five Models

Figure 10 shows the per-layer Cohen’s d on failing vs. succeeding examples for each model (§4.2). Shaded bands mark each model’s activation-patching causal window. Effect sizes peak in the deep layers, downstream of every adapter window tested in the fixed-harness sweep (§5).

I Intrinsic Capacity Metrics: Llama-3 and Mistral

Figure 11 shows C3 (weight effective rank) and C4 (gradient norm) for Llama-3 and Mistral. Both models show the opposite sign pattern from Phi-3.5 (numerical correlations in Table 6). These curves are descriptive: C3/C4 was demoted to post-hoc only after the prospective validation on Qwen-7B was falsified by the fixed-harness layer sweep (§4.4, Table 7).

J Cascade Disruption: Remaining Panel Models

Figure 12 shows the Phi-3.5 cascade disruption profiles referenced in §4.5; Figure 13 shows Llama-3, Mistral, and Qwen2.5-7B. All profiles are measured teacher-forced on shared clean inputs, using the independently retrained sweep adapters of Appendix K. Gemma-2-9B is omitted (adapter-insensitive, §5.2; its original adapters were unavailable for re-measurement).

K Clean Disruption vs. Perturbed Δ : Independent Replication

Figure 14 plots, for the four GSM8K sweep models (Phi-3.5, Qwen2.5-7B, Llama-3, Mistral), total clean-input disruption (cf. Figure 12 and Figure 13) against mean perturbed Δ . Both axes are measured on a single set of *independently retrained* sweep adapters (three seeds per window, fixed harness, max_new_tokens=768; the original §5 checkpoints were no longer available for disruption measurement). Higher disruption tracks lower Δ in

all four models (Spearman $\rho = -0.66 / -0.54 / -0.60 / -0.43$ for Phi/Qwen/Llama/Mistral; $n=6$ windows per model, so no single model reaches significance, but the direction is consistent in all four).

The retraining doubles as a replication of the layer sweep on fresh adapters: the qualitative structure of Table 2 recurs—mid-layer windows worst (Phi L10–14 -5.9 , Qwen L08–11 -7.6 , Llama L10–14 -12.2 , Mistral -18 to -20 pp), deepest windows least harmful—and it additionally covers Llama-3 and Mistral L10–14, which the original sweep did not train (both strongly negative, as cascade disruption predicts). Exact values differ from Table 2 (fresh training; 768 vs. 512 generation budget), and the near-zero late cells fluctuate in sign across retraining (e.g., Qwen L24–27 $+0.3 \rightarrow -0.3$ pp), as expected of effectively null cells. Gemma-2-9B is omitted (adapter-insensitive; §5.2).

L Cross-Task LRD Profiles: Phi-3.5 and Mistral

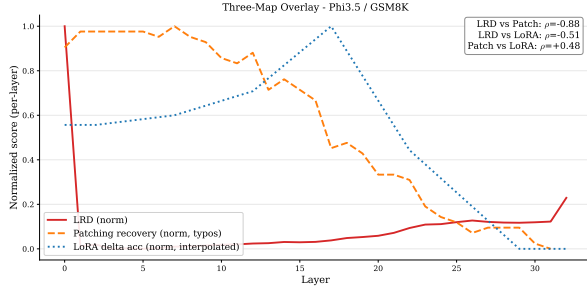
Figure 15 shows the cross-task LRD comparison for the two models on which we collected GSM8K/MMLU/BBH data (§4.6). Both models give $\rho > 0.95$ between tasks, indicating that task modulates LRD magnitude but not profile shape.

M Per-Model Layer Sweep Panels

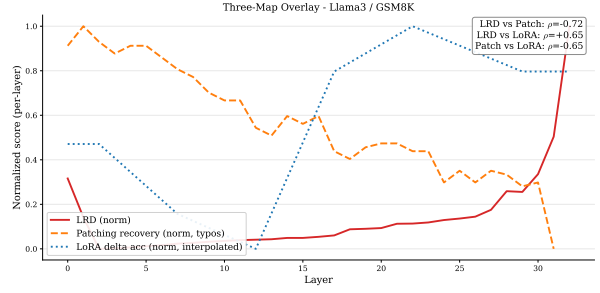
Figure 16 plots the panel-level sweep summarized in Table 2; Figure 17 shows, for each model in the panel, the fixed-harness CE-only LoRA layer sweep on GSM8K broken out by window with three-seed dispersion visible per cell.

The three adjudicable models (Phi-3.5, Qwen2.5-7B, Llama-3) share a clear mid-layer wall (every window in the L05–L20 range is strongly negative), and on each the deepest available window is the least-regressive. Mistral is uniformly negative at the shared learning rate (lr artifact; §5.2); Gemma-2-9B is adapter-insensitive across every window we tested.

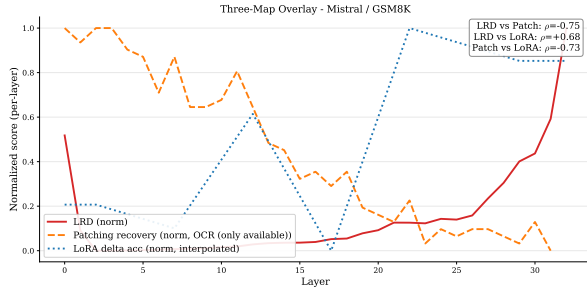
Exact windows behind the nominal columns of Table 2: Phi-3.5 and Llama-3 use L00–04, L05–09, L15–19 (Mid), L20–24 (Mid-late), and L27–31 (Late), with Phi-3.5 adding L10–14 (Early-mid); Llama-3’s grid included no L10–14 window. Qwen2.5-7B uses L00–04, L05–09, L15–19, L20–23, and L24–27, plus L08–11 (Early-mid)—a pre-registered exploratory cell at the C3/C4-predicted



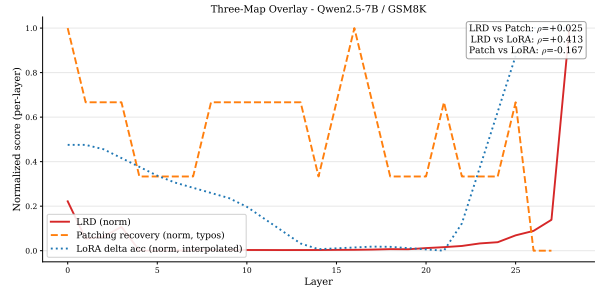
(a) Phi-3.5 (S&S): LoRA peaks where LRD is suppressed.



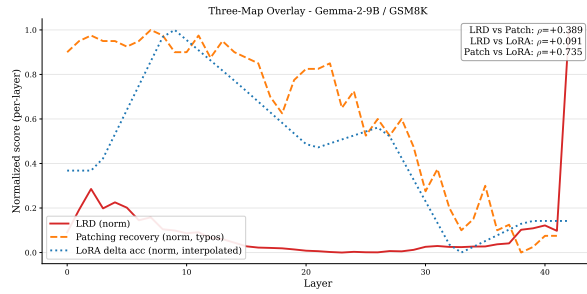
(b) Llama-3 (Late): LoRA peaks where LRD is building.



(c) Mistral (Late): LoRA anti-correlated with patching window.

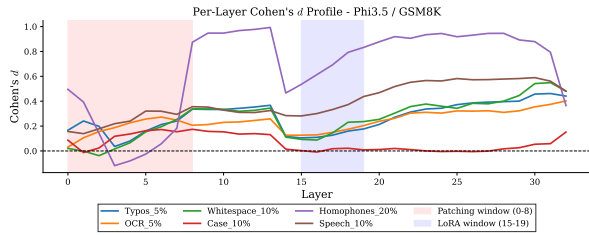


(d) Qwen2.5-7B (Late): LRD $\times$ Patch unusable ($n=3$).

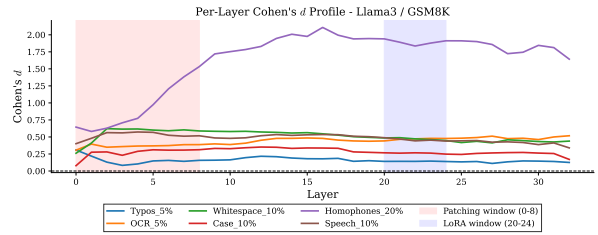


(e) Gemma-2-9B (S&S): exception, $\rho=+0.389$ (60% ceiling).

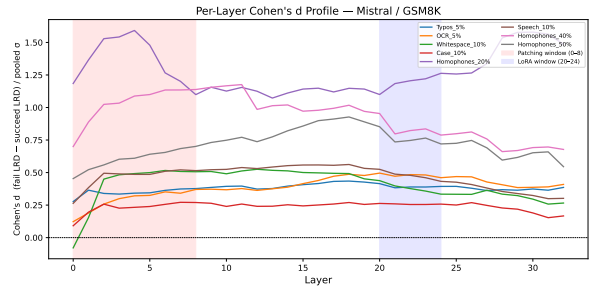
Figure 9: **Three-map overlays, all five models.** LRD sensitivity, patching recovery, and LoRA effectiveness per layer.



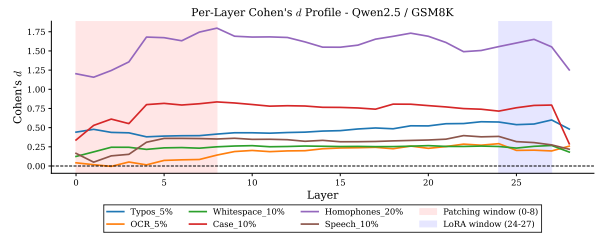
(a) Phi-3.5 (S&S): peak at L28–32.



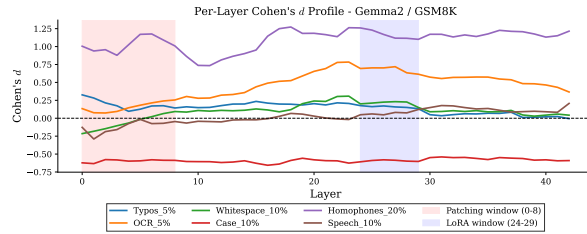
(b) Llama-3 (Late).



(c) Mistral (Late).

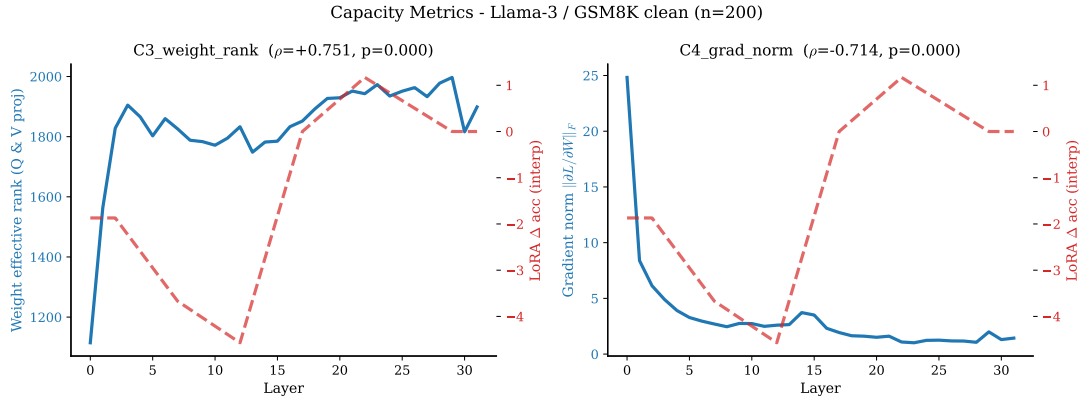


(d) Qwen2.5-7B (Late): peak $d \approx 1.80$ at L8 (Homophones 20%).

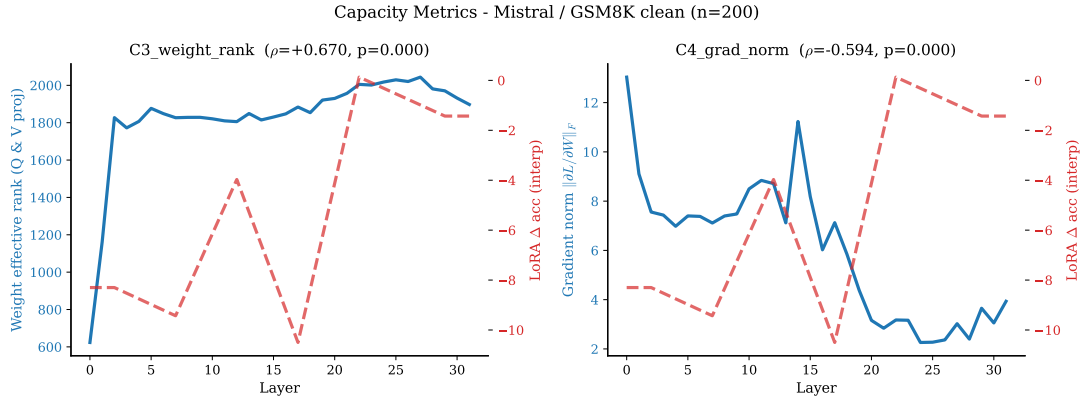


(e) Gemma-2-9B (S&S): peak $d \approx 1.27$ at L18 (Homophones 20%).

Figure 10: **Per-layer Cohen's d , all five models.** Shaded band: activation-patching causal window. Effect sizes peak in the deep layers, downstream of every adapter window tested under the fixed harness.



(a) Llama-3.



(b) Mistral.

Figure 11: Intrinsic Capacity Metrics: Llama-3 and Mistral. C3 (weight effective rank, left) and C4 (gradient norm, right) for each model. Both models show the opposite sign pattern from Phi-3.5 (correlations in Table 6). C1 and C2 correlations are reported in Table 6.

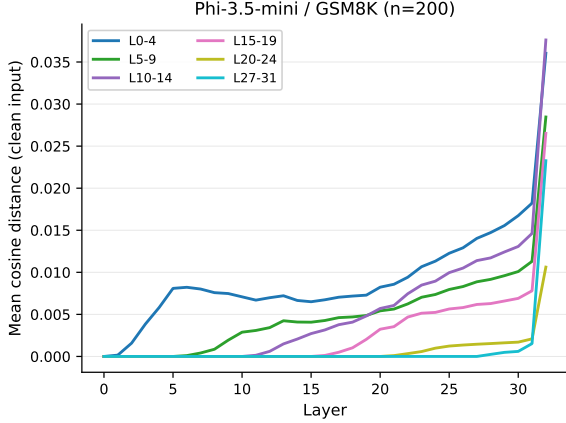


Figure 12: Clean-input disruption for Phi-3.5 (one line per LoRA window; $n=200$). Upstream of each window, divergence is exactly zero; disruption onsets at the window and persists downstream, largest for the earliest windows.

optimum that overlaps the L05–09 partition cell. Gemma-2-9B uses L00–04, L06–11 (reported under the nominal L05–09 column), L15–20, L25–30, and L35–40, single-seed per window (seed-42 for L00–04/L06–11/L15–20; seed-43 for L25–30 and L35–40), with L00–04 replicating at seed-44 (+0.23). Dashes in Table 2 mark cells not trained: Llama-3’s grid included no L10–14 window, Gemma’s coarser five-window grid has no early-mid cell, and all-layer baselines ran only on the three adjudicable models.

N Window-Width Ablation, All-Layer Baseline, and MMLU Control

This section reports the three controls referenced in §5.2: the window-width ablation, the all-layer LoRA baseline, and the MMLU sweep. All use the fixed harness, three seeds per cell, and the setup of §5.

Window-width ablation (Phi-3.5, GSM8K). Table 8 varies the adapted window width at the mid and late positions. The sign structure is width-robust—mid negative at every width, late approximately zero—and mid-window damage grows monotonically with width.

Window position	W3	W5	W7
Mid (L10–14 center)	-2.8 ± 0.6	-5.3 ± 1.1	-7.8 ± 0.4
Late (L27–31 center)	-0.0 ± 0.1	$+0.2 \pm 0.3$	-0.6 ± 0.5

Table 8: Window-width ablation on Phi-3.5 (GSM8K). Mean perturbed Δ (pp) $\pm$ std across three seeds; Wn = window width n . Width-3 windows are L11–13 (mid) and L28–30 (late); width-7 are L09–15 and L25–31 (clamped at the 32-layer boundary). Width-5 values are the original sweep runs (Table 2, max_new_tokens=512); widths 3 and 7 are new runs under the fixed harness at 768.

All-layer LoRA baseline (GSM8K). Table 9 gives the all-layer baseline underlying the last column of Table 2, alongside the clean-accuracy regression: all-layer adaptation damages clean performance on all three adjudicable models, as expected if its early-layer share incurs cascade disruption.

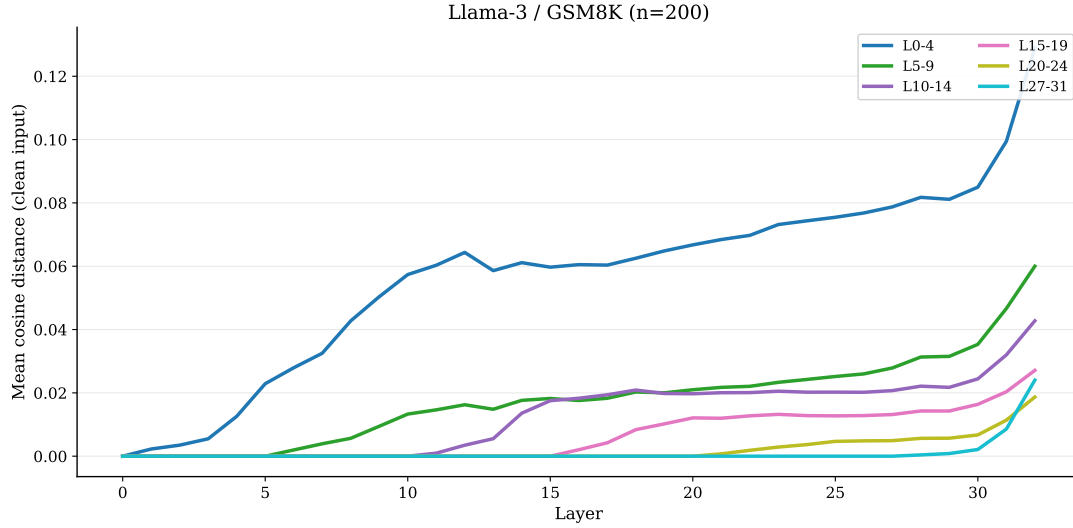
Model	Mean perturbed Δ	Clean Δ
Phi-3.5-mini	-8.6 ± 0.7	-8.1 ± 0.2
Qwen2.5-7B	-13.6 ± 0.1	-14.1 ± 0.3
Llama-3-8B	-9.9 ± 1.0	-11.4 ± 0.7

Table 9: All-layer LoRA baseline (GSM8K, fixed harness). Mean Δ (pp) $\pm$ std across three seeds, relative to the no-adaptor baseline. Qwen’s tight std reflects distinct checkpoints whose six-condition means nearly coincide (per-condition accuracies differ by up to 3 pp across seeds); all-layer adaptation leaves no seed-dependent window choice to vary.

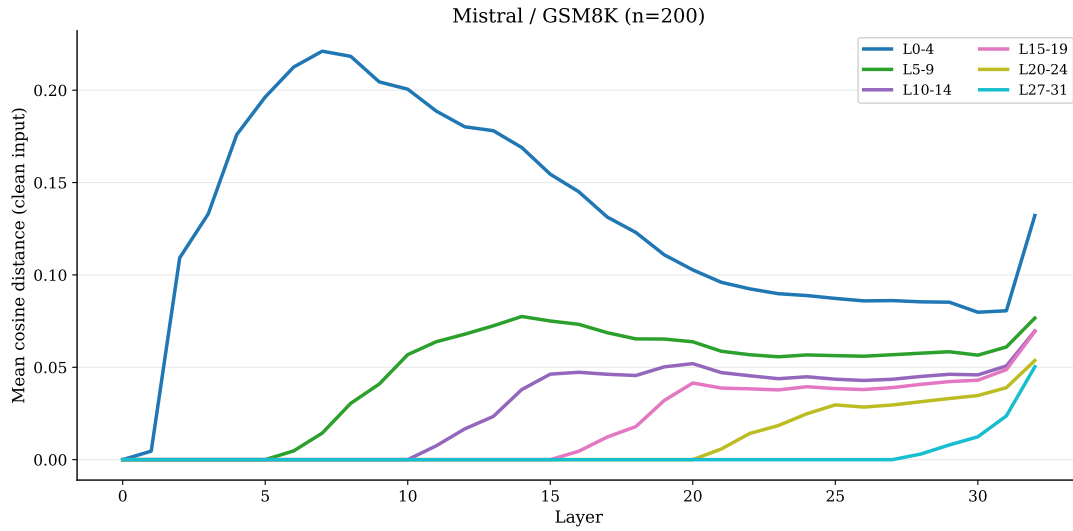
MMLU control sweep. Table 10 reports the fixed-harness sweep on MMLU for one model per regime. Effects are near-zero throughout; the only reliably nonzero cell is Phi-3.5’s diagnostic-flagged mid window L10–14 (-1.8 ± 0.4 pp). No window produces positive transfer on either model. See §5.2 for interpretation; we do not claim the GSM8K mid-layer wall replicates on MMLU.

Phi-3.5-mini		Qwen2.5-7B	
Window	Δ (MMLU)	Window	Δ (MMLU)
L00–04	$+0.4 \pm 0.2$	L00–04	$+0.0 \pm 0.1$
L05–09	$+0.4 \pm 0.1$	L05–09	$+0.1 \pm 0.0$
L10–14	-1.8 ± 0.4	L08–11	-0.0 ± 0.1
L15–19	-0.4 ± 0.3	L15–19	$+0.0 \pm 0.1$
L20–24	-0.3 ± 0.1	L20–23	$+0.1 \pm 0.1$
L27–31	-0.0 ± 0.1	L24–27	-0.1 ± 0.1

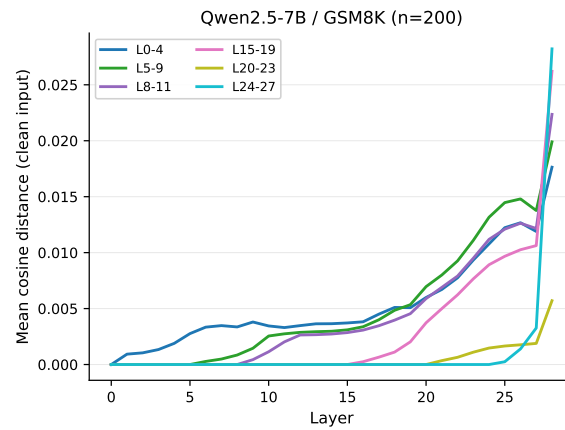
Table 10: MMLU layer sweep (fixed harness, one model per regime). Mean perturbed Δ (pp) $\pm$ std across three seeds per window.



(a) Llama-3.

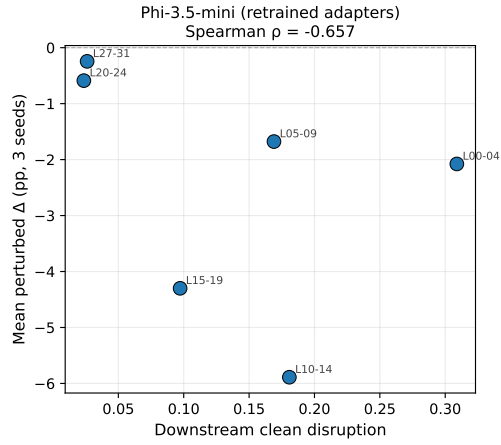


(b) Mistral.

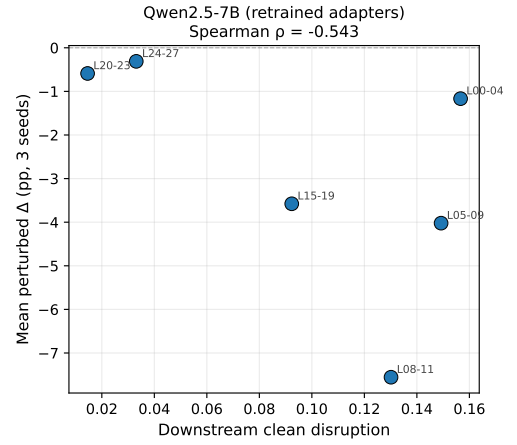


(c) Qwen2.5-7B.

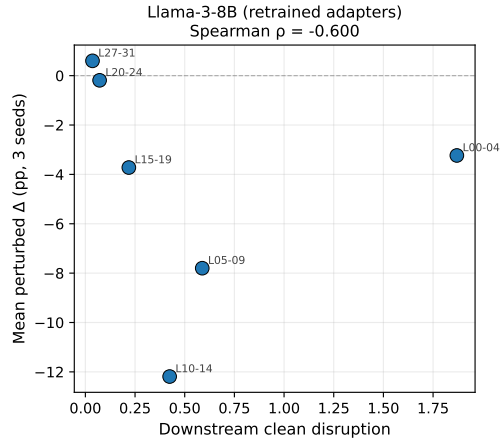
Figure 13: **Cascade Disruption: Remaining Panel Models.** Clean-input disruption (one line per LoRA window; $n=200$; independently retrained adapters, cf. Appendix K). On all three models, divergence is exactly zero upstream of the adapted window, onsets at the window, and persists through every downstream layer; total disruption is largest for the earliest windows and shrinks with window depth, consistent with Phi-3.5 (Figure 12).



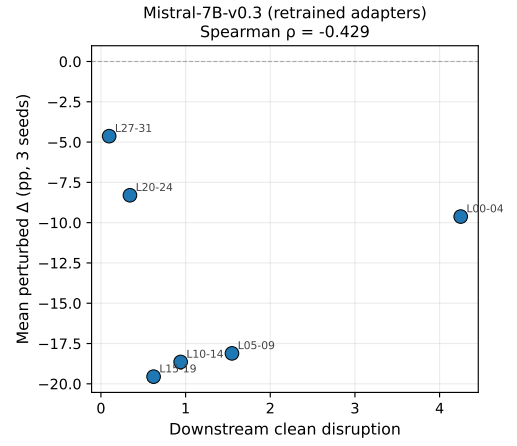
(a) Phi-3.5.



(b) Qwen2.5-7B.

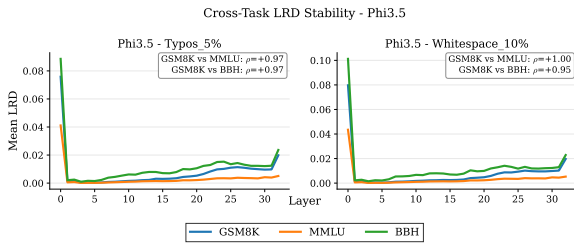


(c) Llama-3.

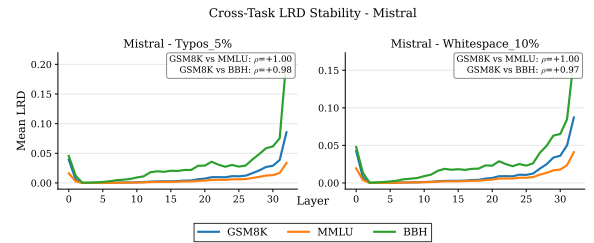


(d) Mistral.

Figure 14: Total clean-input disruption vs. mean perturbed Δ on independently retrained sweep adapters (both axes from the same checkpoints; three seeds per window). Direction is consistent in all four models.



(a) Phi-3.5 (spike-and-suppress).



(b) Mistral (late-accumulation).

Figure 15: Cross-task LRD profiles (GSM8K, MMLU, BBH; typos 5%) for Phi-3.5 and Mistral. Both produce near-identical shapes across tasks ($\rho > 0.95$).

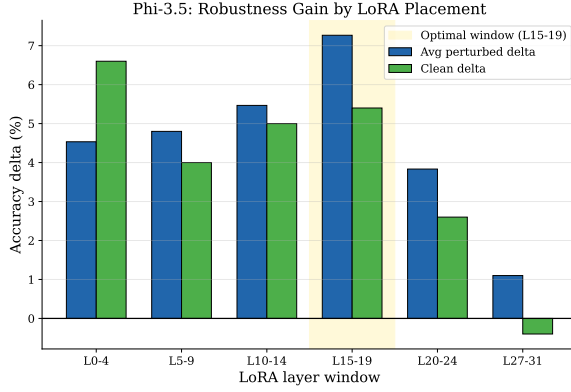


Figure 16: Fixed-harness CE-only LoRA layer sweep on GSM8K. Mean perturbed Δ (six perturbations, three seeds) on Phi-3.5, Qwen2.5-7B, Llama-3, and Mistral. On the three adjudicable models the mid-layer L08–L20 range is uniformly most damaging; non-negative windows are the deepest available. Mistral’s row reflects an lr artifact (§5.2). Gemma-2-9B is adapter-insensitive (± 0.7 pp) and reported separately.

O Earlier Investigations (Pre-Fixed-Harness)

This section catalogs investigations carried out before the evaluation harness bug was identified (§5.3). All numerical results in those experiments were generated by an evaluation harness with `max_new_tokens=100`, which truncated chain-of-thought generations before the `####` NUMBER extraction line and scored truncated answers as empty. The same checkpoints read between +10 and +20 pp differently under the fixed harness, depending on condition (see Figure 18 and §5.3). The qualitative conclusions of these investigations are preserved below; full numerical tables are omitted from this revision because they are not trustworthy on absolute scale.

Cosine versus MSE stability loss, with matched-rate control. We ran cosine and MSE variants of the representation-stability loss at the original variable per-perturbation rates and at a matched 10% rate. At the original rates, a per-condition split appeared (cosine winning on typos/OCR/speech/homophones, MSE on whitespaces/cases), which we had provisionally interpreted as a taxonomy: “directional” perturbation types suited to a cosine loss versus “uniform” types suited to MSE. The matched-rate control collapsed the split: Cohen’s d values for the two groups overlapped completely (0.40–0.48 vs. 0.15–0.48), and the pooled interaction

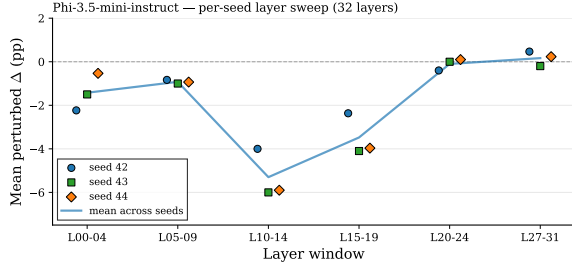
CI went from $[+2.5\%, +8.7\%]$ at variable rates to $[-2.5\%, +3.5\%]$ at matched rates. The taxonomy claim from this loss-geometry work was therefore disproved by the matched-rate control; the underlying intervention has now also been disproved by the harness fix.

λ_{stab} scaling sanity check. At $\lambda_{\text{stab}} = 0.3$, L_{stab} contributed $\leq 0.6\%$ of the total loss in every run. Setting $\lambda_{\text{stab}} = 3.0$ raised the contribution to 6–8%. This established that earlier cosine-vs-MSE comparisons at $\lambda_{\text{stab}} = 0.3$ were essentially noise. Under the fixed-harness ablation reported in the main paper, however, even the properly weighted $\lambda_{\text{stab}} = 1.0$ regime *hurts* CE LoRA by 1.2–3.9 pp on the conditions and seeds we tested. The stability loss is not a useful component once the evaluation harness is correct.

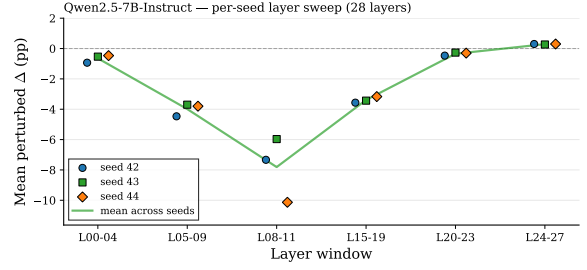
LoRA capacity experiments. We swept LoRA ranks $\{16, 32, 64\}$ at $\{3,000, 5,000\}$ training steps on Llama-3 and Mistral and observed monotonic degradation with rank and steps. The qualitative conclusion—that the prior stabilizer’s failure on late-accumulation models is not a capacity issue—survives the harness fix. A separate “capacity concentration” result claimed that a concentrated 5-layer LoRA at L15–L19 outperforms an all-layer LoRA by +5.1 pp perturbed; that +5.1 number is broken-harness and is excluded from this revision. The cascade-disruption explanation in the main paper does not depend on it.

Transfer evaluation. GSM8K-trained adapters with the prior stability loss degraded MMLU by 1.5–4.5 pp and were near-zero on BBH. Task-specific MMLU training produced near-zero average gains. Diagnostic-side cross-task stability (LRD profile $\rho > 0.95$ across GSM8K/MMLU/BBH on Phi-3.5 and Mistral) is harness-independent and is retained in the main paper.

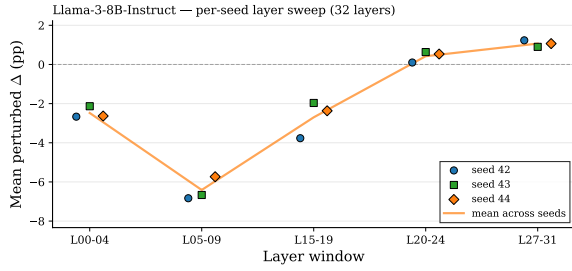
Alternative residual-adapter stabilizer. Independently of the LoRA-based stabilizer, we tested a lightweight two-layer bottleneck residual adapter $x \mapsto x + g \cdot W_2 \sigma(W_1 x)$ inserted after the attention block at four placement configurations on Qwen-7B (layers 2/6/10/14; 8/12/16/20; 12/16/20/24; 4/10/18/26) and at the first eight layers on Mistral. Training mixed clean and OCR-perturbed inputs with clean targets. Every configuration regressed: Qwen-7B baseline outperformed all four stabilized variants by approximately 9–11 pp on GSM8K



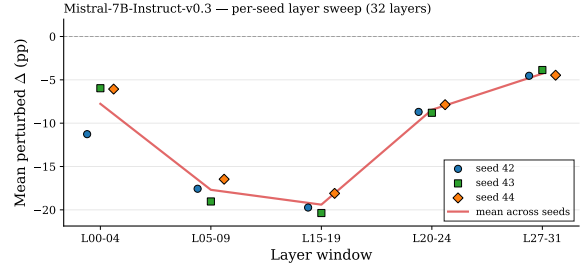
(a) Phi-3.5 (spike-and-suppress; clean 85.4%).



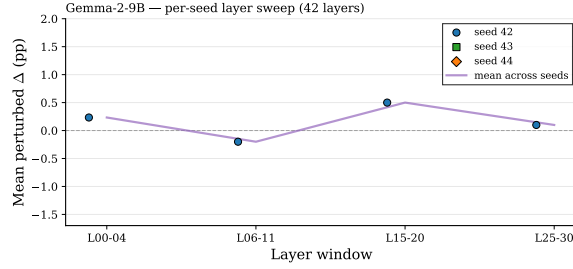
(b) Qwen2.5-7B (late-accumulation; clean 89.0%).



(c) Llama-3-8B (late-accumulation; clean 78.4%).



(d) Mistral-7B-v0.3 (late-accumulation; clean 59.6%; panel outlier).



(e) Gemma-2-9B base (spike-and-suppress; clean 64.8%; single-seed per window with one cross-seed replication at L00-04).

Figure 17: **Per-Model Layer Sweep Panels.** Fixed-harness CE-only LoRA, mean perturbed Δ per non-overlapping 5-layer window, three seeds per window (seeds 42, 43, 44). The numerical summary is in Table 2.

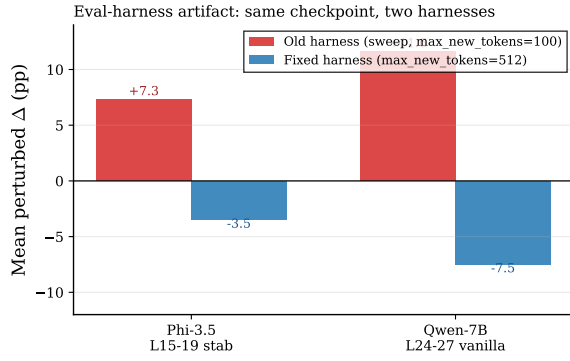


Figure 18: Same checkpoints, two harnesses (§5.3). Mean perturbed Δ for Phi-3.5 L15–19 with the stability loss (earlier harness: +7.3 pp) and the Qwen2.5-7B L24–27 vanilla CE baseline (earlier harness: +11.6 pp), under the earlier `max_new_tokens=100` harness (red) vs. the fixed harness (blue); $n=500$. The fixed harness uses `max_new_tokens=512` for this checkpoint comparison and the layer sweep (Table 2); per-experiment settings in Table 11. The earlier gains were truncated chain-of-thought scored as empty.

OCR-5%; Mistral baseline exceeded the stabilized variant by approximately 4.5 pp. The qualitative result (multiple stabilizer designs, including a non-LoRA one, fail to improve robustness on these models) is consistent with the main-paper negative result. As with the rest of this section, the absolute numbers are pre-fixed-harness.

Full per-condition numbers from each of these investigations are available on request and will be released with the public code drop.

P Reproducibility Settings

Component	Configuration
Eval harness	CoT, ### extraction
max_new_tokens	512: layer sweep (Table 2), two-harness comparison
	768: diagnostics, scaling, width-3/7, all-layer
	32: MMLU
Diagnostics sample size	$n = 500$ per condition
Scale-variation sample size	$n = 500$ test; $n_{\text{clean-correct}}=200$
Patching pairs	Phi-3.5 $n=50$; Llama-3 $n=100$; Qwen-14B $n=84/100$
LoRA rank / alpha	4 / 8
LoRA targets	q_proj, v_proj
LoRA loss	cross-entropy only ($\lambda_{\text{stab}}=0$)
Layer-sweep windows	non-overlapping 5-layer windows across depth
Layer-sweep seeds	{42, 43, 44} (three per window)
Training steps	300 (Phi-3.5); 1,000–3,000 (Llama-3, Mistral, Qwen, Gemma)
Bootstrap	block size 5, 10,000 resamples
Hardware	NVIDIA A40 (46GB), A100 (80GB), V100 (32GB); single-GPU jobs
Cluster	SLURM
Total compute	≈900 GPU-hours (reported experiments; excludes exploratory and failed runs)

Table 11: Reproducibility settings for the fixed-harness experiments in the main paper. The $\lambda_{\text{stab}}=1.0$ ablation referenced in §5.3 uses the same settings with the stability-loss term re-enabled. The pre-fixed-harness investigations in §O used different settings (`max_new_tokens=100`, $\lambda_{\text{stab}}=3.0$, fixed optimal windows).

MMLU control sweep. The MMLU sweep (§5.2, Table 10) uses `max_new_tokens=32` and a

fixed evaluation subset; the item IDs for that subset are included in the released code.

Software stack. Single-GPU PyTorch 2.10.0+cu128 with the following libraries: transformers 4.44.2, peft 0.11.0, datasets 4.5.0, accelerate 1.12.0, numpy 2.2.5, scipy 1.15.3. Exact pinned versions and the full requirements.txt accompany the code release.

Model checkpoints and loading. Weights are loaded from the official HuggingFace releases: microsoft/Phi-3.5-mini-instruct, mistralai/Mistral-7B-Instruct-v0.3, meta-llama/Meta-Llama-3-8B-Instruct, google/gemma-2-9b, Qwen/Qwen2.5-7B-Instruct, Qwen/Qwen2.5-1.5B-Instruct, Qwen/Qwen2.5-14B-Instruct, and meta-llama/Llama-3.2-1B-Instruct. All models load in `torch_dtype=bfloat16` with `attn_implementation="eager"` where required by the activation-patching hooks.

LoRA configuration (peft). `r=4`, `lora_alpha=8`, `target_modules=["q_proj", "v_proj"]` for Mistral, Llama-3, Gemma-2, and all Qwen scales; `target_modules=["qkv_proj", "o_proj"]` for Phi-3.5-mini (fused attention). Inference-mode flag toggled per call; otherwise default peft arguments.

Datasets (datasets). openai/gsm8k (main config, test split) drawn with shuffle seed 42 and `n_samples` $\in \{200, 500\}$ per condition as reported in Table 11; MMLU and BBH loaded from their respective official HuggingFace releases (cited in §4).

Optimizer (torch). `torch.optim.AdamW` with default $(\beta_1, \beta_2, \epsilon)$ and weight decay; learning rates as specified in the main paper (shared-grid sweep at $5e-5$, Mistral audit at $1e-5$). Training in `bfloat16` except where noted.

Statistics (scipy, numpy). `scipy.stats.pointbiserialr`, `scipy.stats.mannwhitneyu`, `scipy.stats.levene`, and `scipy.stats.spearmanr` are used at default arguments for the per-perturbation significance checks and the pairwise correlations reported

above. The moving-block bootstrap is an in-house NumPy implementation (block size 5, 10k resamples).